Prerequisite Lab

Setting Up SageMaker Domain and Launching Jupyter Notebooks in SageMaker Studio

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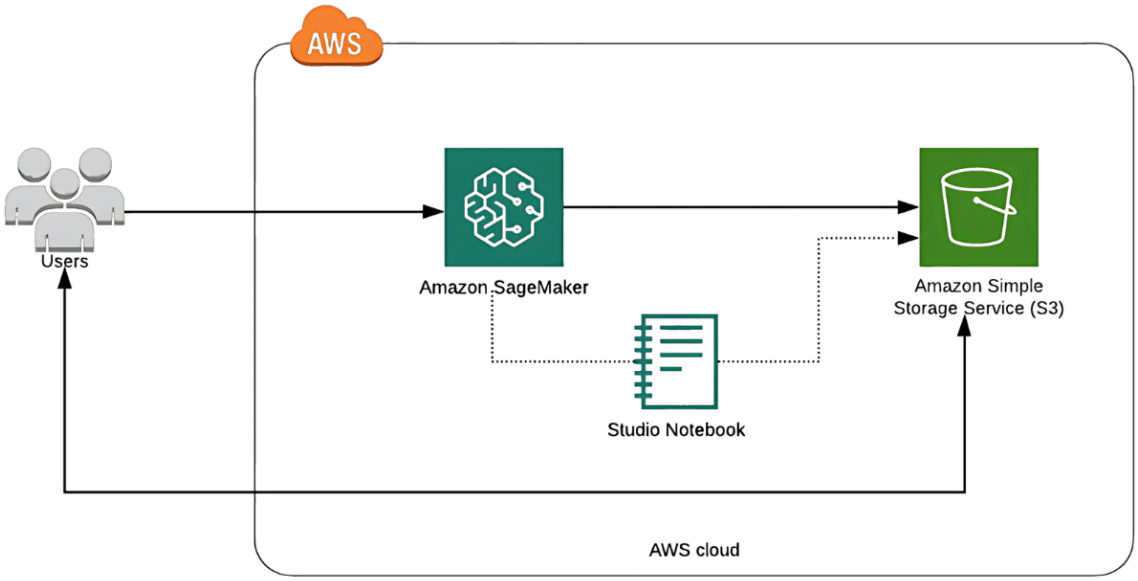
1 INTRODUCTION

Amazon SageMaker Studio is an **integrated development environment (IDE)** tailored for **machine learning (ML) tasks.**

It provides a seamless interface for managing your ML workflows, including data preprocessing, model training, and deployment.

Configuring Jupyter Notebook in SageMaker Studio is an essential step for running Python code, training models, and performing data analysis.

Jupyter notebooks provide an interactive way to write, test, and visualize code results.



In this guide, we will walk you through the steps to configure and use Jupyter Notebook in Amazon SageMaker Studio for your machine learning projects.

This lab will guide you through:

- **Creating a SageMaker Studio Domain:** Setting up the foundation for your AI/ML development.
- **Configuring Jupyter Notebooks:** Launching a workspace and optimizing it for coding and data analysis.

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2 Overview and Key Concepts

In this section, we'll cover key concepts used in this guide, including an overview of what's covered in this guide.

2.1 What is SageMaker Studio

SageMaker Studio is an integrated development environment (IDE) that AWS provides specifically for machine learning (ML) tasks. It offers a web-based interface where you can build, train, and deploy machine learning models all in one place.

Key features of SageMaker Studio include:

- **Jupyter Notebooks**: You can write, run, and debug code in Jupyter Notebooks directly within SageMaker Studio.
- **Collaboration**: Easily share projects, code, and data with team members, making it great for collaborative work.
- **Resource Management**: Manage compute resources like CPU and GPU instances from the interface, scaling up or down as needed.
- **Integrated Tools**: Access tools for data wrangling, model monitoring, and debugging, all without leaving the Studio environment.

2.2 What is Jupyter Notebook

Jupyter Notebook is an open-source web application that allows you to create and share documents containing live code, equations, visualizations, and narrative text. Jupyter Notebook serves as a key tool for developing and testing machine learning models.

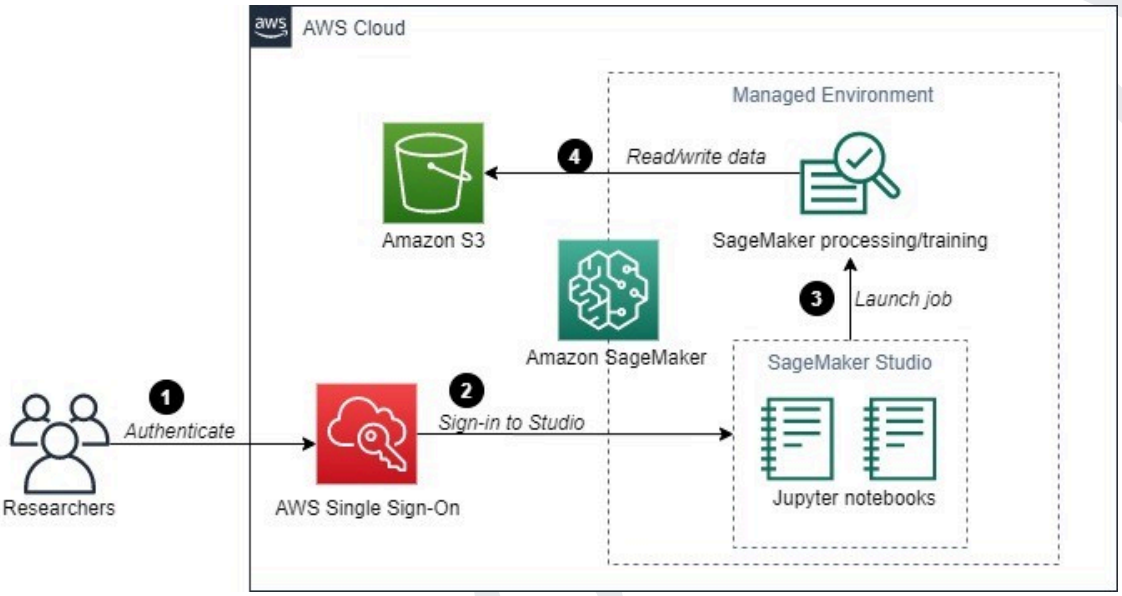
In SageMaker Studio, Jupyter Notebooks offer:

- **Interactive Development:** Execute code in real-time and view the output immediately within the notebook interface.
- **Data Exploration and Visualization:** Easily explore datasets and visualize data with integrated plotting libraries.
- **Modular Workflows:** Break down complex workflows into individual cells, allowing for incremental development and testing.

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- **Seamless Integration:** Jupyter Notebooks within SageMaker Studio are tightly integrated with other AWS services, making it easy to access data stored in S3, run training jobs, and deploy models.

  Using Jupyter Notebooks in SageMaker Studio, you can efficiently prototype, iterate, and refine your machine-learning models in a collaborative environment.



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3 Documentation Links

1. **Amazon SageMaker:**
   https://docs.aws.amazon.com/sagemaker/latest/dg/whatis.html

2. **SageMaker Studio:**
   https://docs.aws.amazon.com/sagemaker/latest/dg/studio.html

3. **AWS Machine Learning:**
   https://aws.amazon.com/machine-learning/

4. **Jupyter Notebook**
   https://docs.jupyter.org/en/latest/

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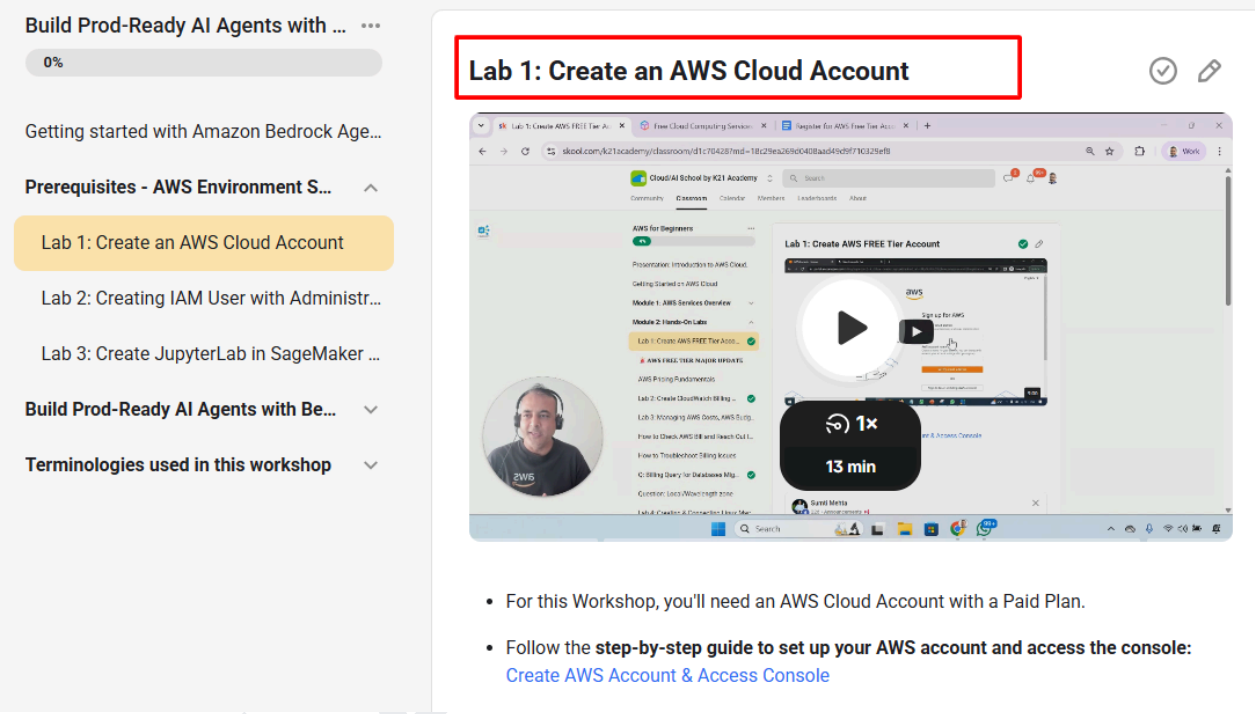
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4 PREREQUISITES

4.1 Create an AWS Cloud Account

1. For this Workshop, you'll need an AWS Cloud Account with a Paid Plan.

2. Follow the **step-by-step guide to set up your AWS account and access the console:** Create AWS Account & Access Console

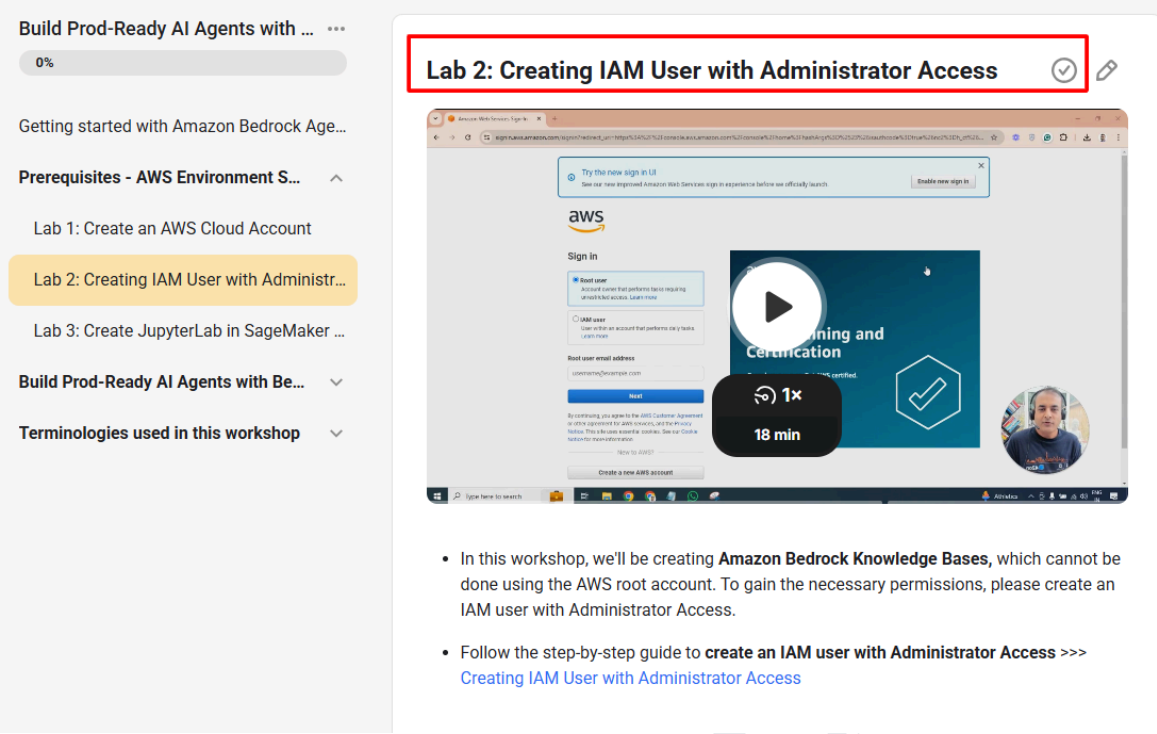


4.2 Creating IAM User with Administrator Access

1. In this workshop, we'll be creating **Amazon Bedrock Knowledge Bases,** which cannot be done using the AWS root account. To gain the necessary permissions, please create an IAM user with Administrator Access.

2. Follow the step-by-step guide to **create an IAM user with Administrator Access** >>> Creating IAM User with Administrator Access

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- In this workshop, we'll be creating **Amazon Bedrock Knowledge Bases,** which cannot be done using the AWS root account. To gain the necessary permissions, please create an IAM user with Administrator Access.
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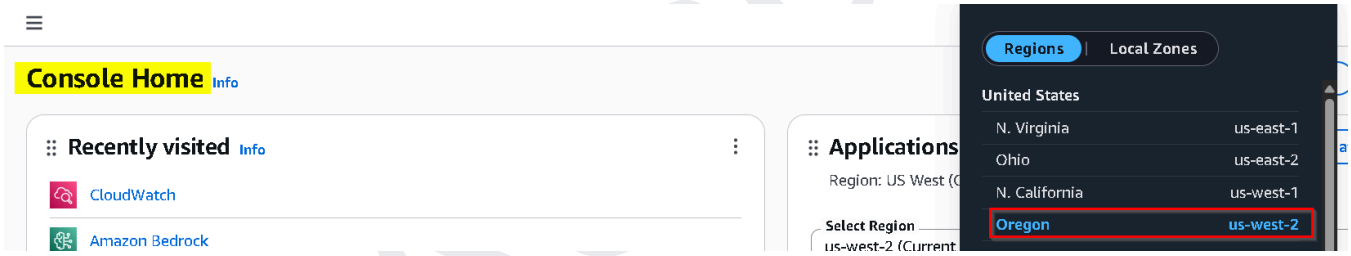
5 Create an Amazon SageMaker Domain

Creating a **SageMaker Domain** sets up the foundational environment that allows you and your team to access and manage SageMaker Studio resources securely. It acts as the workspace hub for running notebooks, training models, and collaborating.

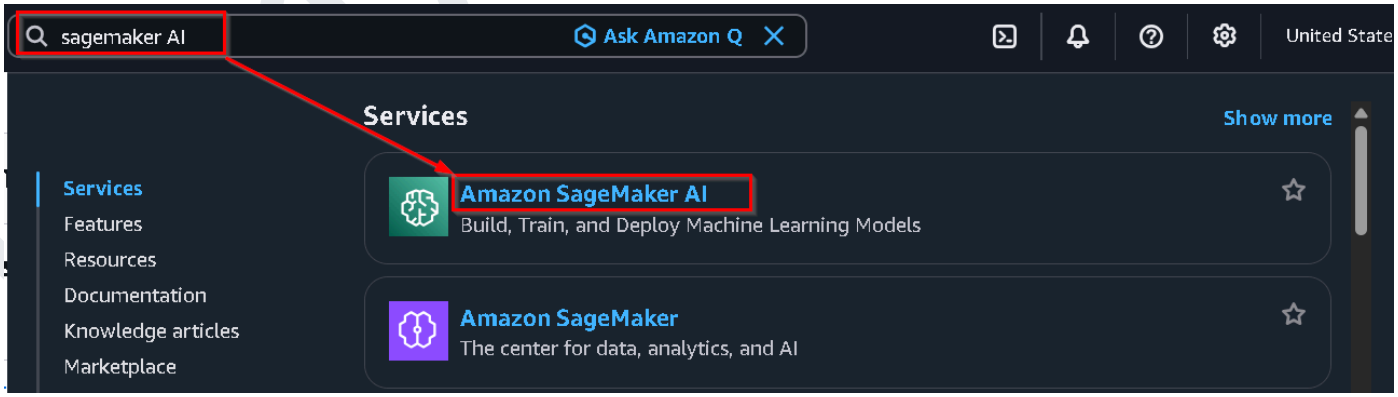
In this section, we will create a SageMaker Domain using the AWS Console by following the quick setup process. This includes creating a domain and a user profile so you can launch and use SageMaker Studio seamlessly.

Note: Before starting the lab, please ensure you have reviewed and completed all the requirements listed in the prerequisites section to avoid any issues during the lab.

1. Log in to the AWS Console and select the **us-west-2** region.

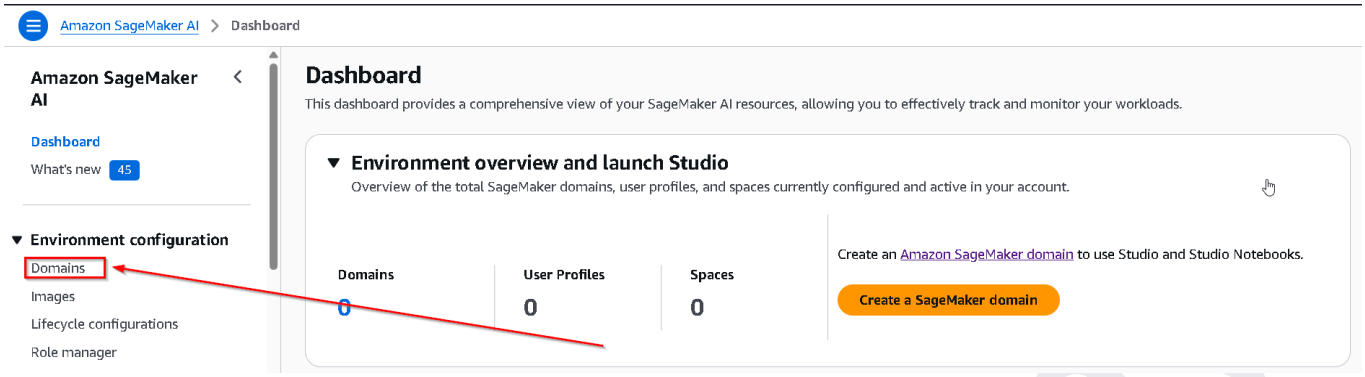


2. Search for **SageMaker AI** and open the service.

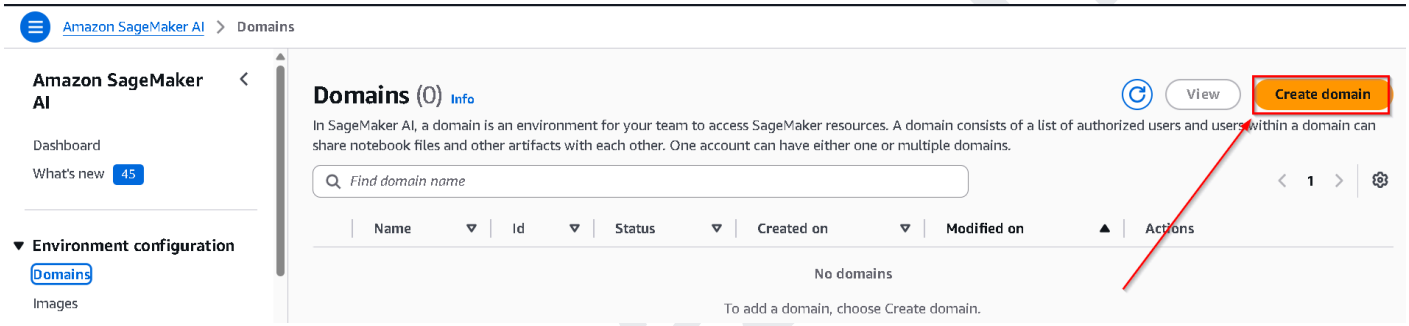


3. In the left menu, click **Domains**.

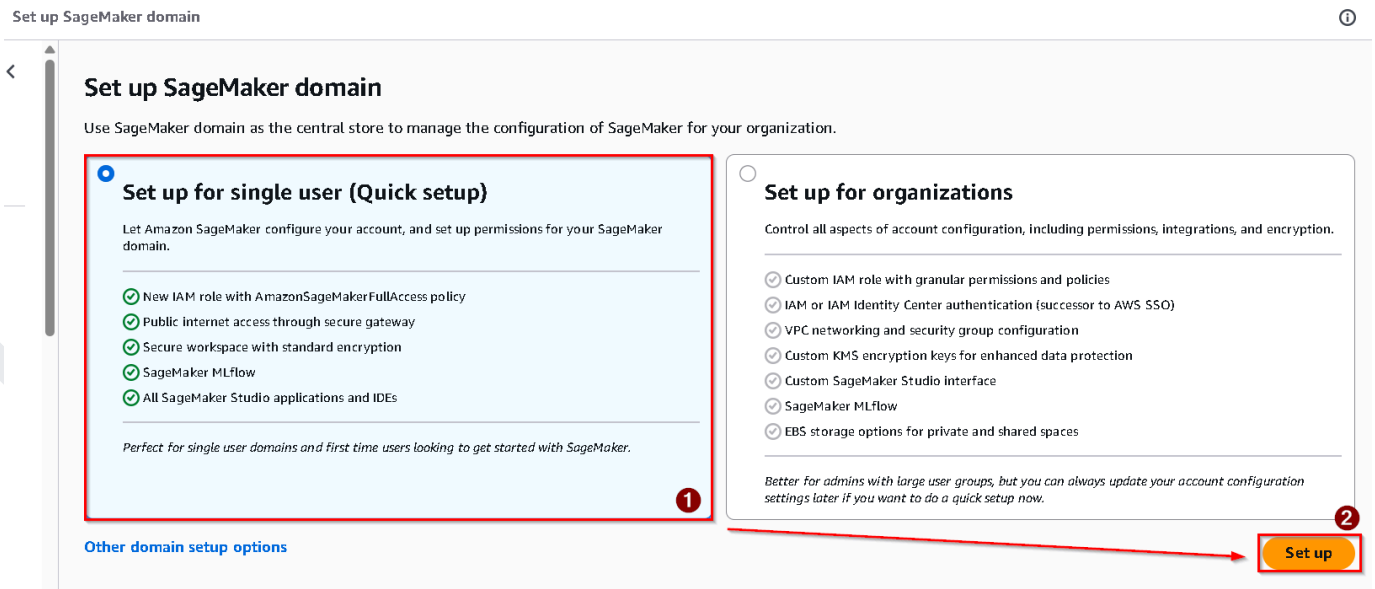
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4. Click **Create domain**.



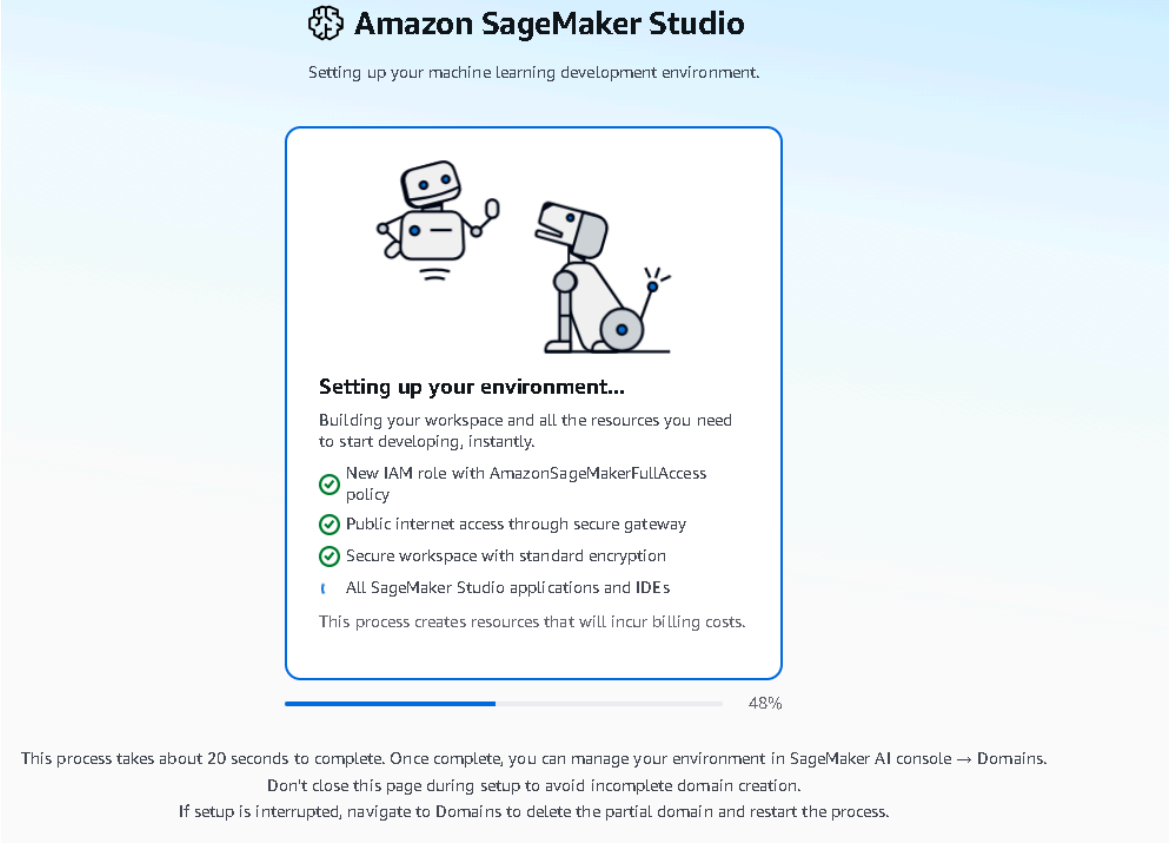
5. Choose "**Set up for single user (Quick setup)**." This will automatically create a domain and user profile for you.



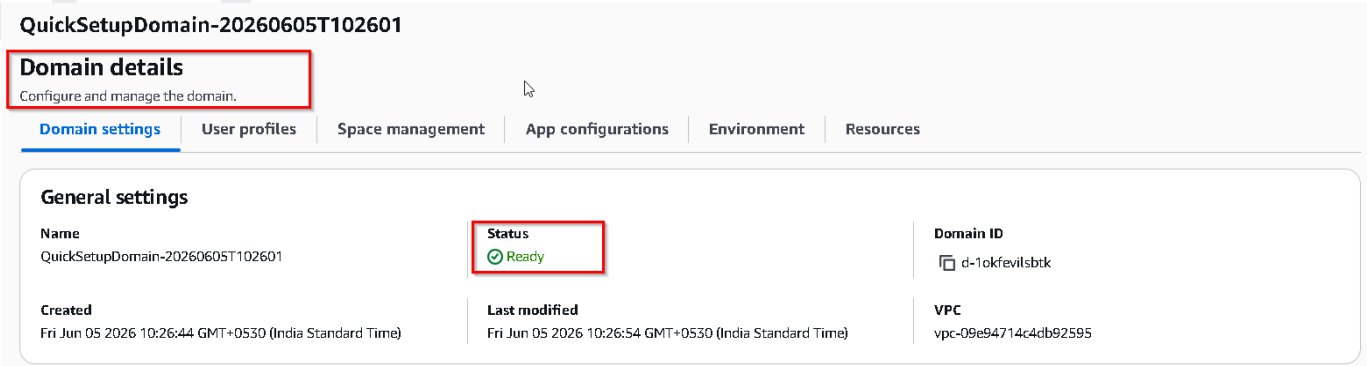
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6. Once the quick setup begins, you will see the message:
   "Setting up your machine learning development environment"

- Please be patient and avoid closing the tab until the setup finishes.



- The quick setup automatically creates a SageMaker domain and user profile using default settings. **As an administrator, you can view and modify these domains and user profiles as needed.**



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The Quick Setup is now complete. You can begin using SageMaker Studio and explore its features.

5.1 Update the default configurations in the SageMaker Domain

In this workshop, *Build Prod-Ready AI Agents with Bedrock AgentCore*, we will utilize various AWS services like S3, CodeBuild, Lambda, Cognito, Bedrock, and AgentCore. To ensure full functionality and smooth operation of your SageMaker Domain, specific configurations need to be updated.

These configurations may include:

1. **Administrator Access**: To ensure SageMaker Domain can interact with various AWS services such as S3, CodeBuild, Lambda, Cognito, Bedrock, and AgentCore, you have to assign **Administrator Access** to the SageMaker Domain. This will allow SageMaker to perform required tasks such as creating resources, managing permissions, and facilitating integration with other services.

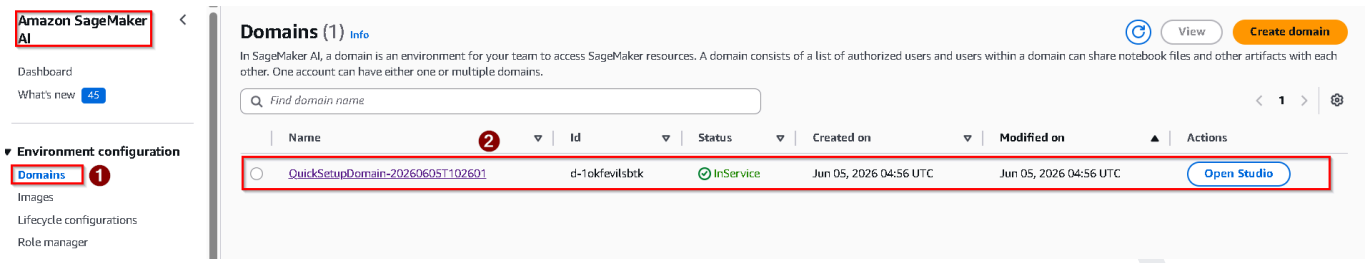
2. **Enable Docker in SageMaker Domain**: For this workshop, we will require Docker support in SageMaker to handle containerized environments. You will need to enable Docker within the **App Configurations** of the SageMaker Domain. This will allow seamless execution of Dockerized applications during the training and deployment phases of your AI agents.

By following these steps, your SageMaker Domain will be properly configured to handle all the components required for building production-ready AI agents with Bedrock AgentCore.

5.1.1 Attach "AdministratorAccess" Policy to SageMaker Domain

1. In the Amazon SageMaker AI service, click on **Domain** and select the domain you created.

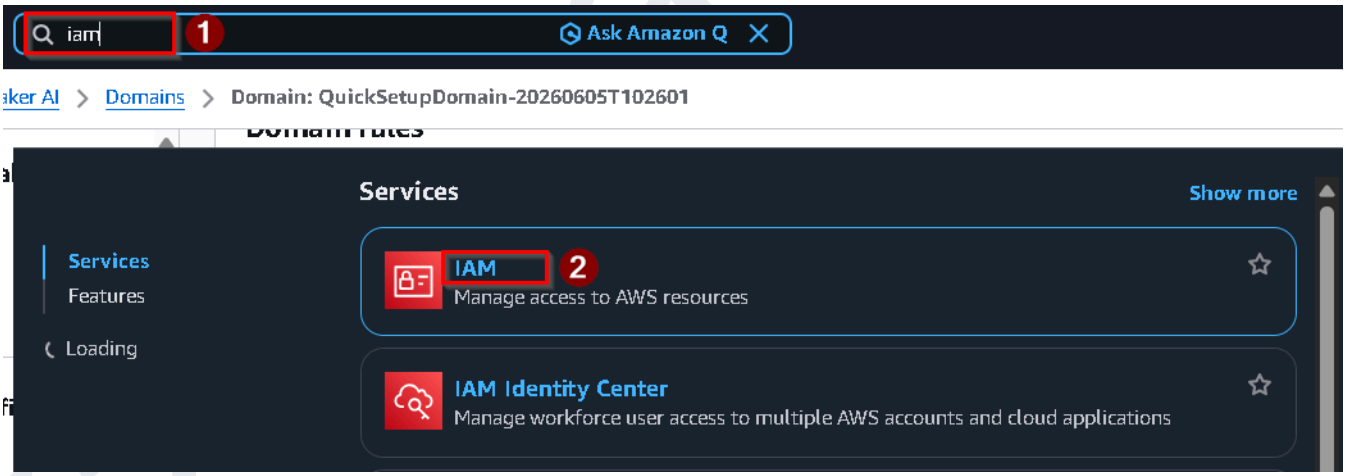
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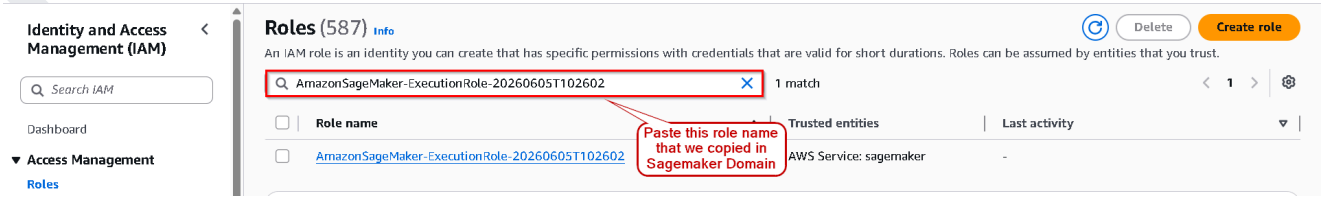
2. In **Domain Settings** > **Authentication and Permissions**, copy the role name as shown in the image below.



3. In the AWS Console, search for **IAM** and select the **IAM** service.

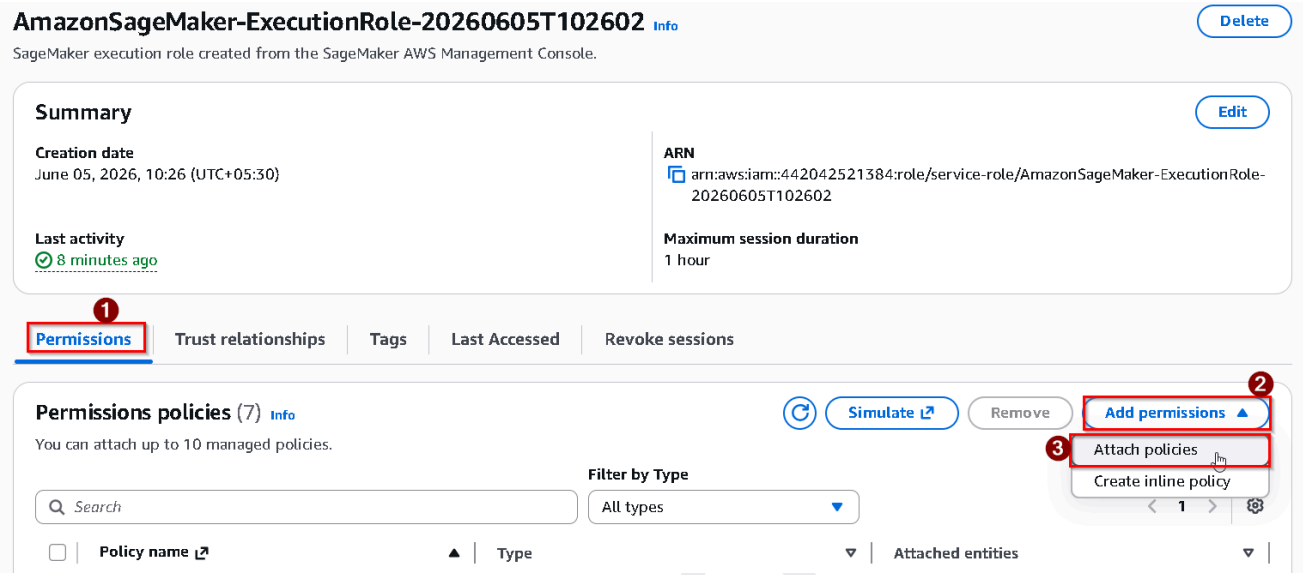


4. In the **IAM** section, click on **Roles**. Paste the role name you copied from Amazon SageMaker Domain into the search bar, then select the role to add the policy.

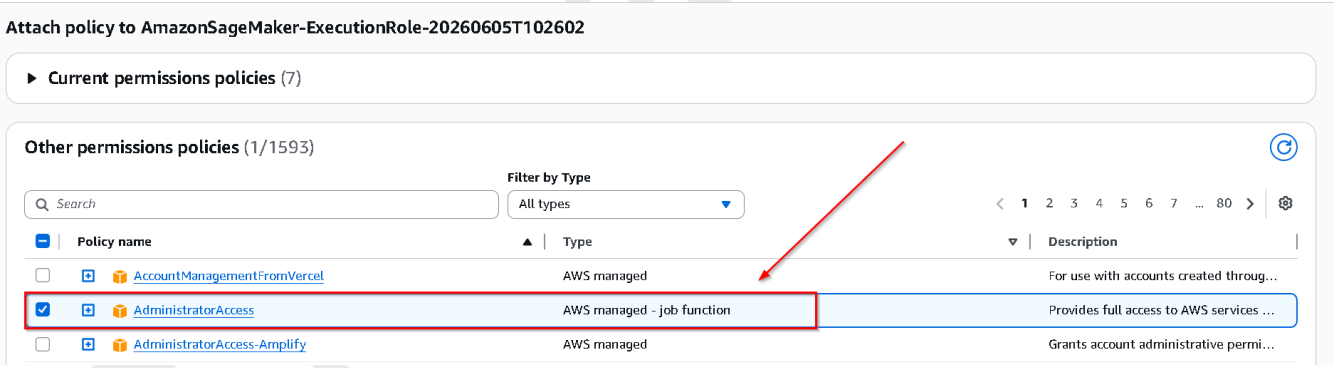


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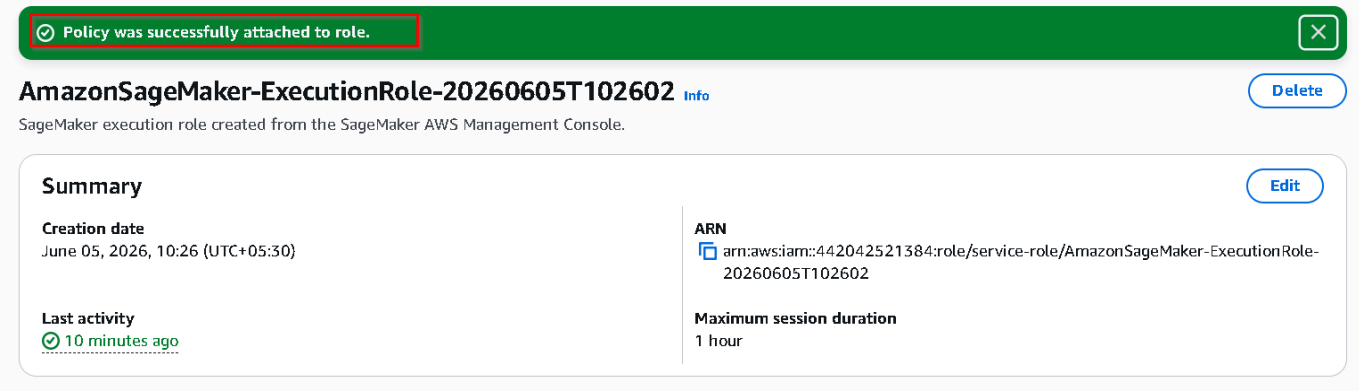
5. In the **Permissions** section, click on **Add Permissions** and then select **Attach Policies**.



6. Select the **AdministratorAccess** policy, and then click on **Add Permissions**.

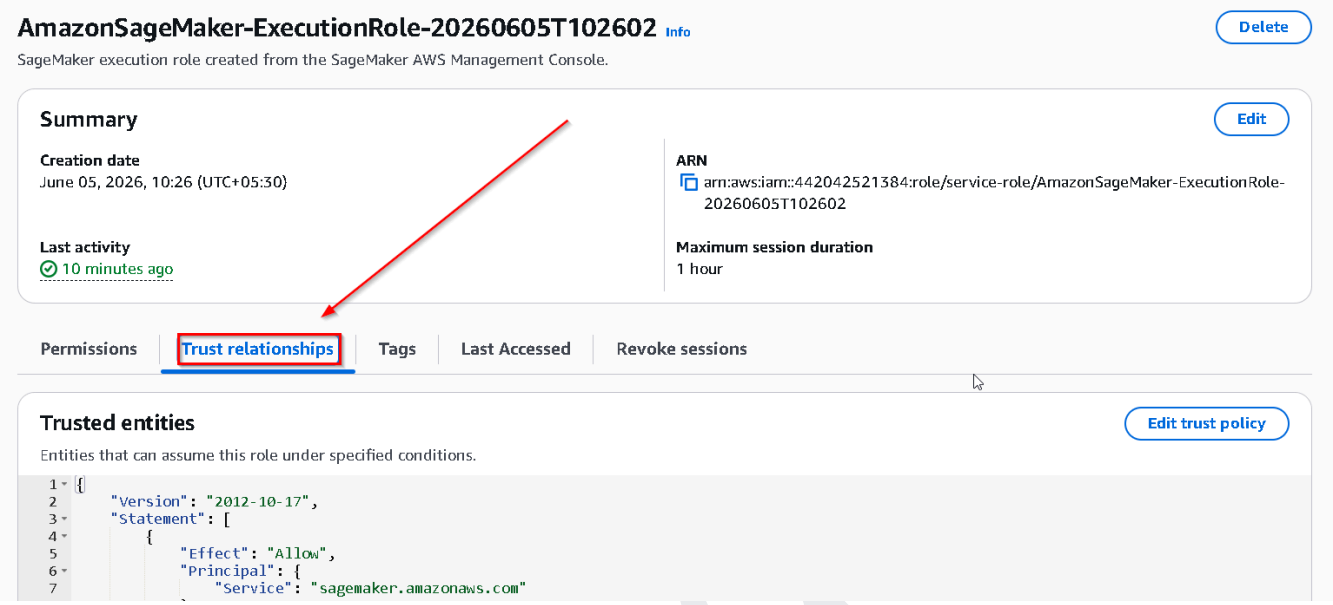


7. The policy was successfully attached to the role.

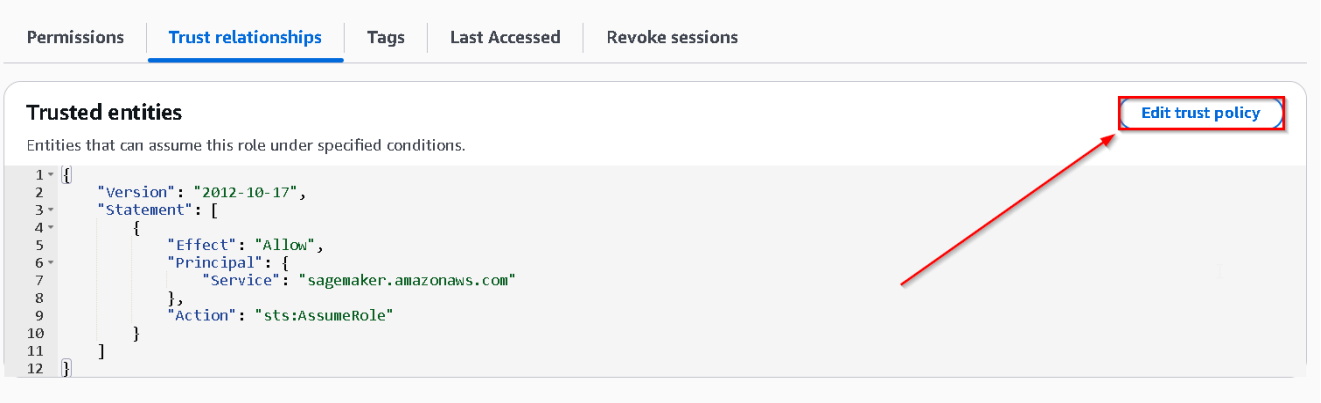


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8. After adding the Administrator Access policy, click on **Trust Relationships** to configure the trust settings for the IAM role.



9. Click on the **Edit Trust Policy**.



10. Copy and paste the code below into the trust policy, then click **Update Policy**.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
```

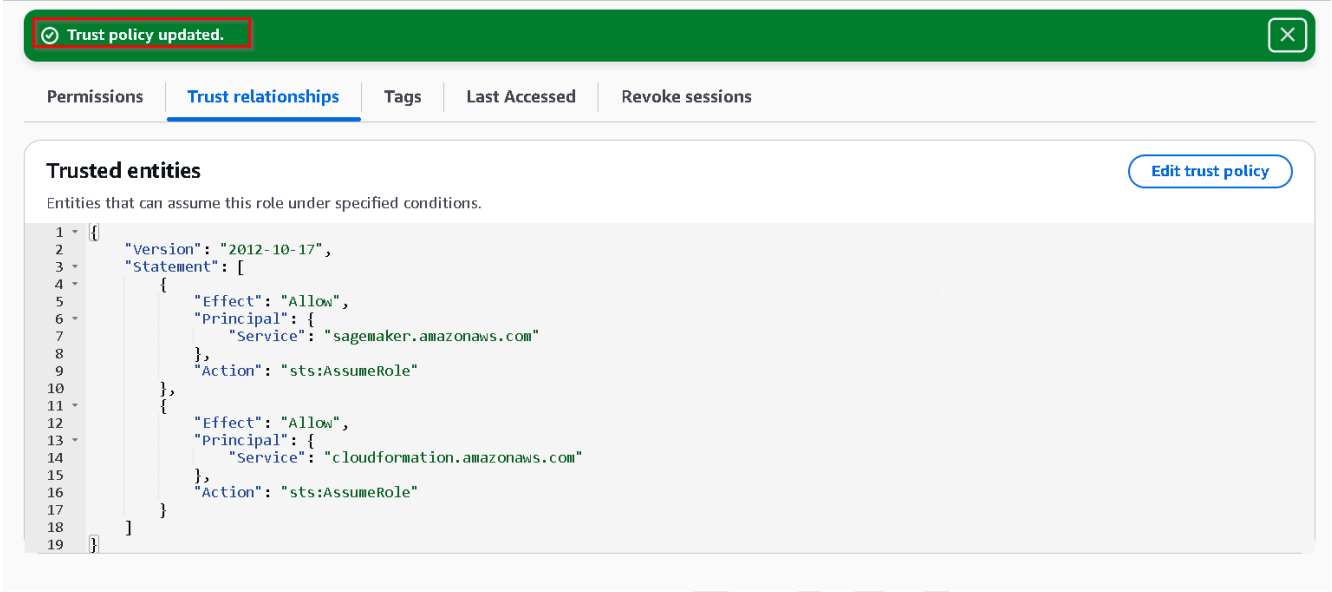
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```
            "Service": "sagemaker.amazonaws.com"
        },
        "Action": "sts:AssumeRole"
    },
    {
        "Effect": "Allow",
        "Principal": {
            "Service": "cloudformation.amazonaws.com"
        },
        "Action": "sts:AssumeRole"
    }
  ]
}
```



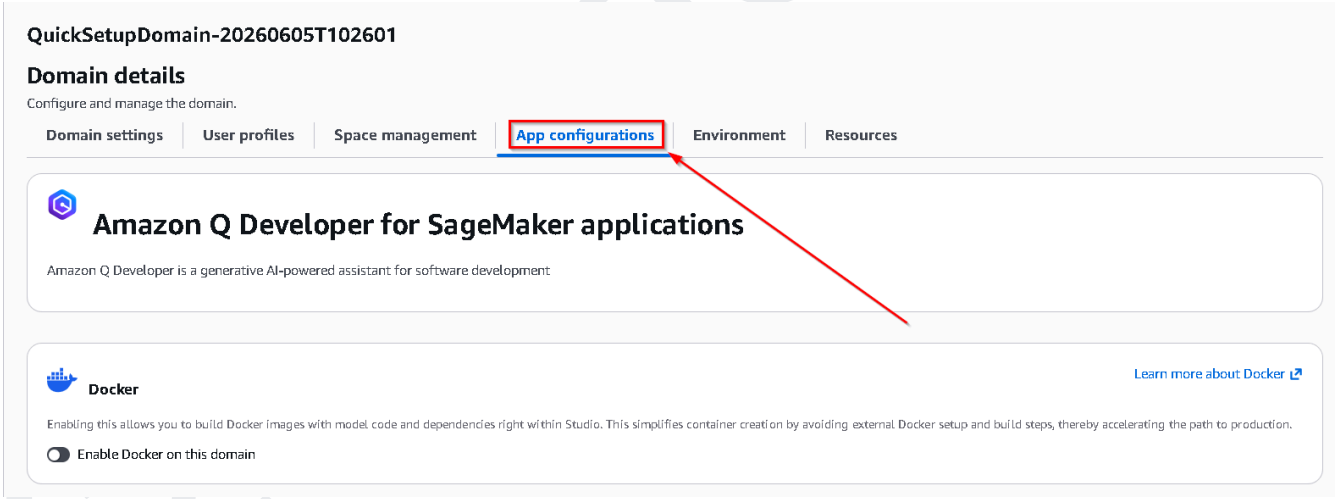
11. Trust policy is now updated.

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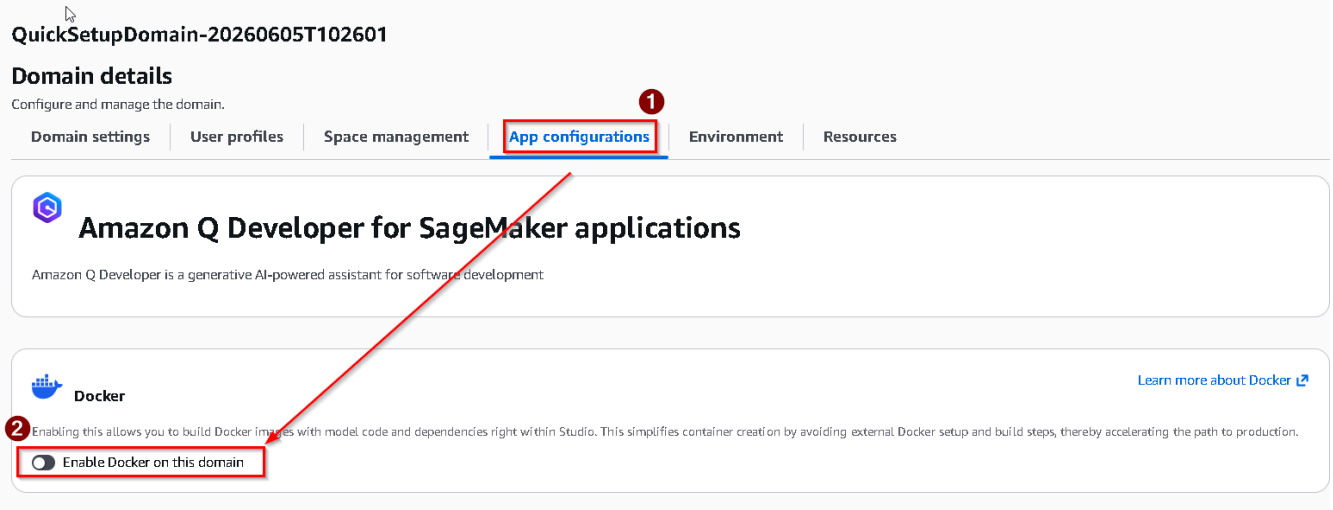
5.1.2 Enable Docker on Sagemaker Domain

1. Go to the SageMaker Domain and select **App Configuration**

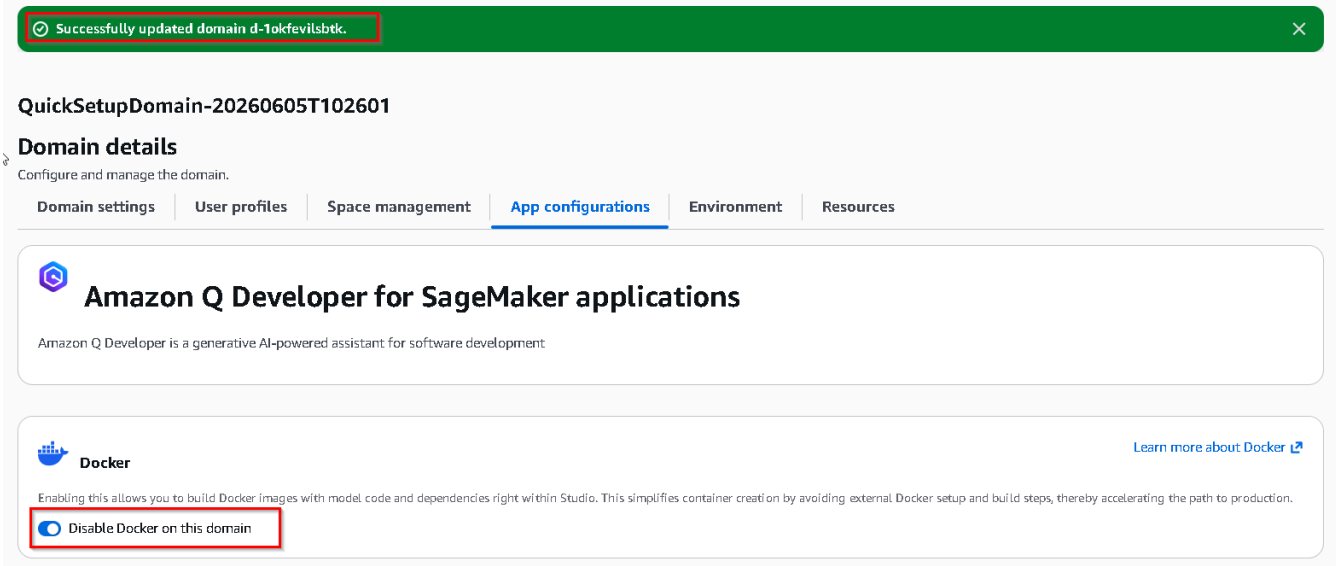


2. Scroll down and navigate to the Docker Configuration section, then enable the option **Enable Docker on this domain**.

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3. Now, Docker is enabled on this domain.



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6 LAUNCH SAGEMAKER STUDIO

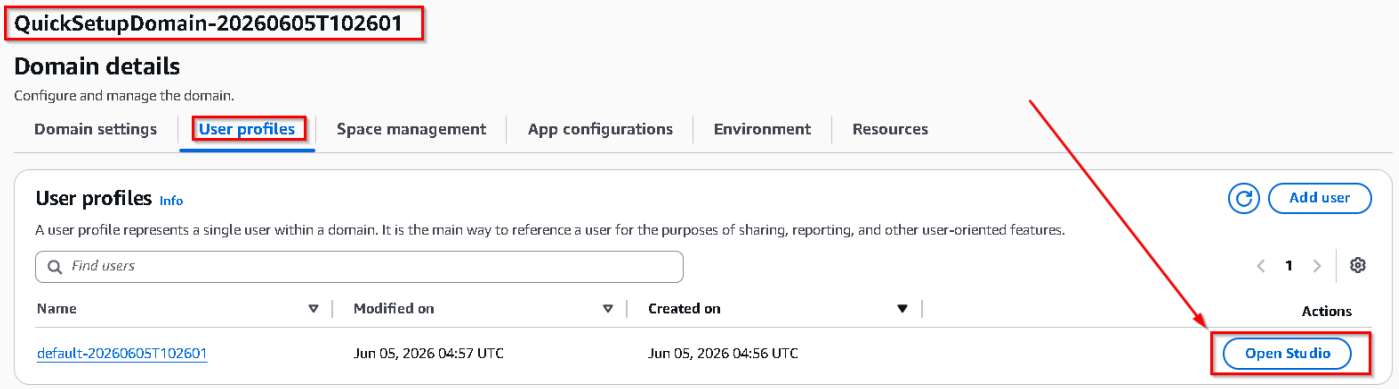
Accessing SageMaker Studio is essential because it provides a powerful, fully integrated web-based environment where you can develop, train, and deploy machine learning models.

This unified interface lets you work efficiently without needing to install or configure software locally.

In this section, you will launch your SageMaker Studio user profile from the AWS console and open the SageMaker Studio integrated development environment (IDE) directly in your browser.

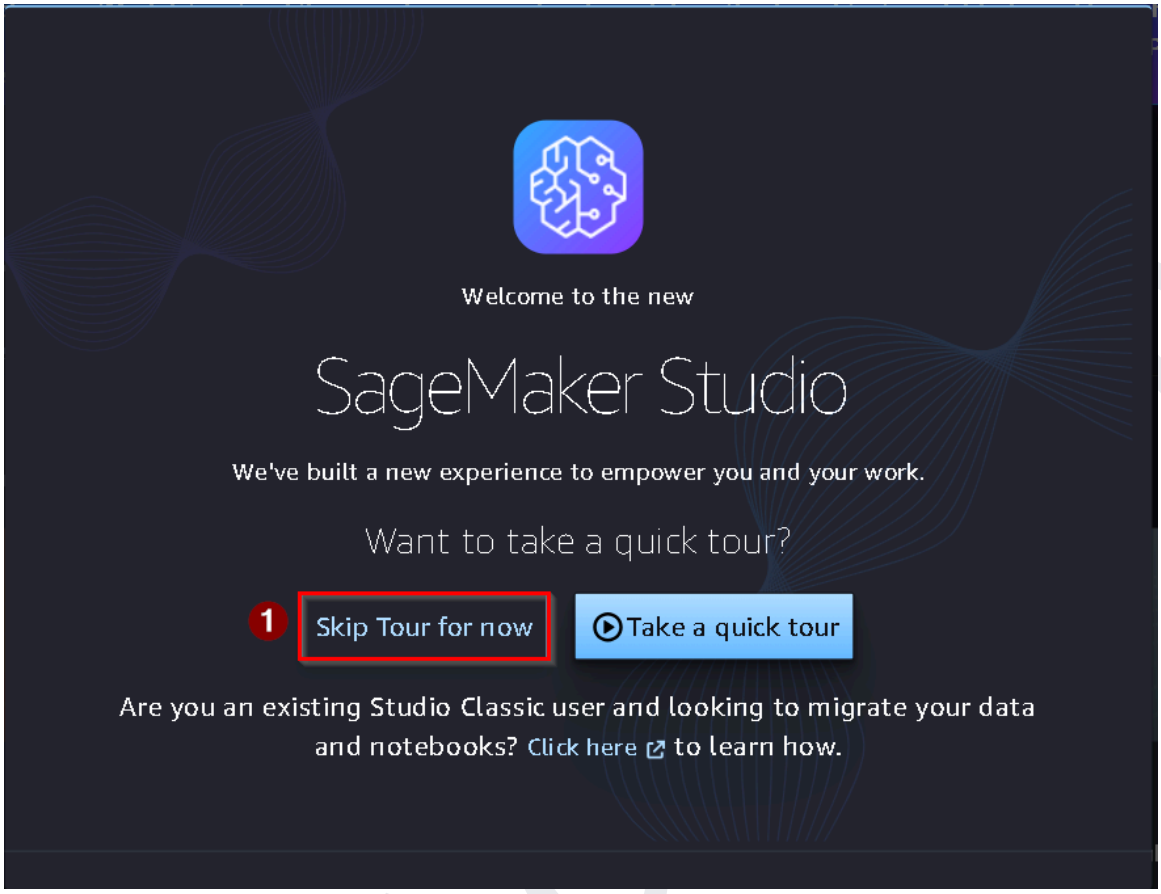
Completing these steps grants you access to the central workspace where all your machine learning development activities, such as coding, training, and deploying models, will take place.

1. Click on the **user profiles** under **Domain details,** then click **Open Studio** to open SageMaker Studio.



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2. When SageMaker Studio opens, choose **Skip Tour**

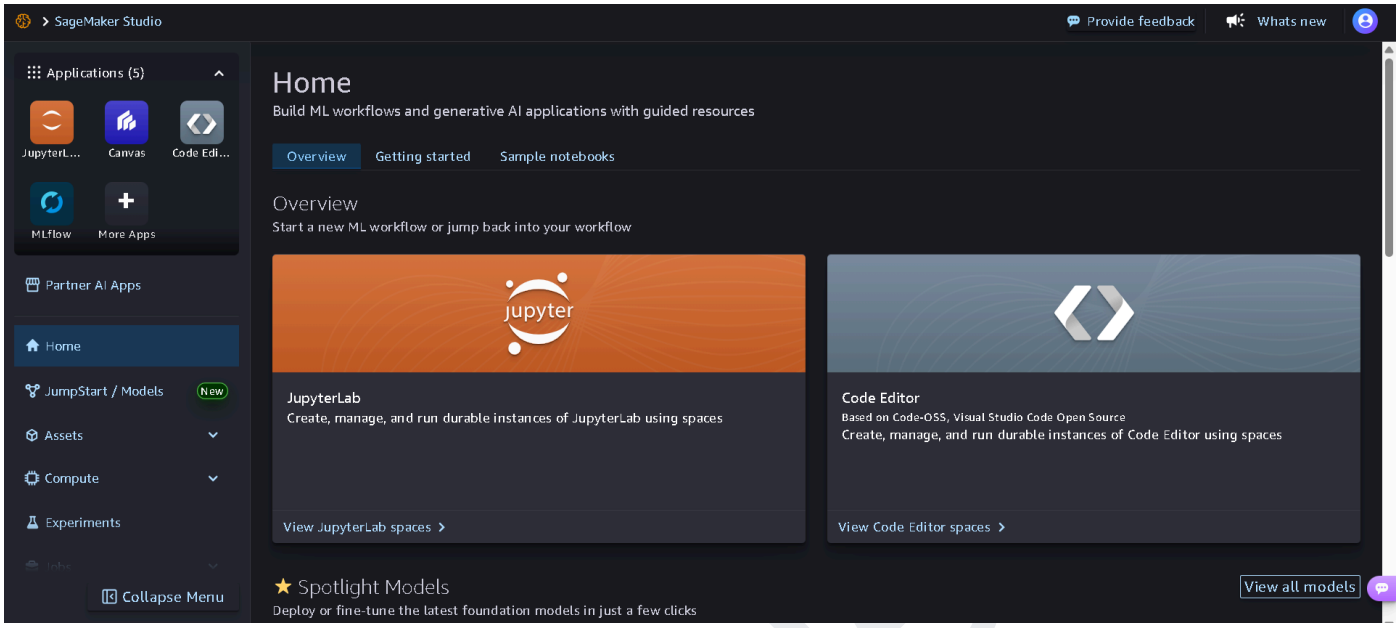


Note: If you want to **explore all the features** of **SageMaker Studio**, you can click **"Take a quick tour"** to get an **overview of its functionalities.**

Note: SageMaker Studio's web interface is your central hub for machine learning development. It allows you to launch your preferred IDE, **such as JupyterLab, RStudio, or a fully managed Code Editor based on Visual Studio Code.**

These tools are accessible within flexible workspaces called Spaces, enabling you to build, train, tune, and deploy models efficiently

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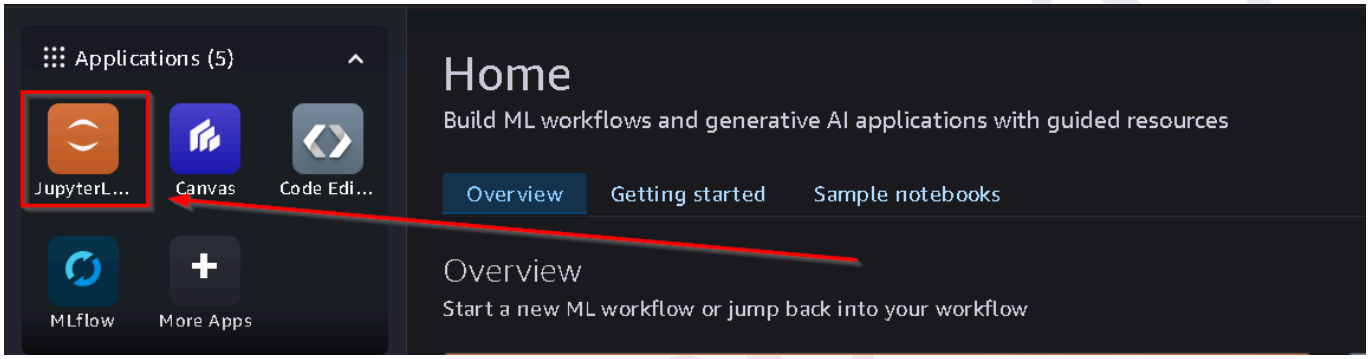
Congratulations! You have successfully launched and accessed SageMaker Studio.

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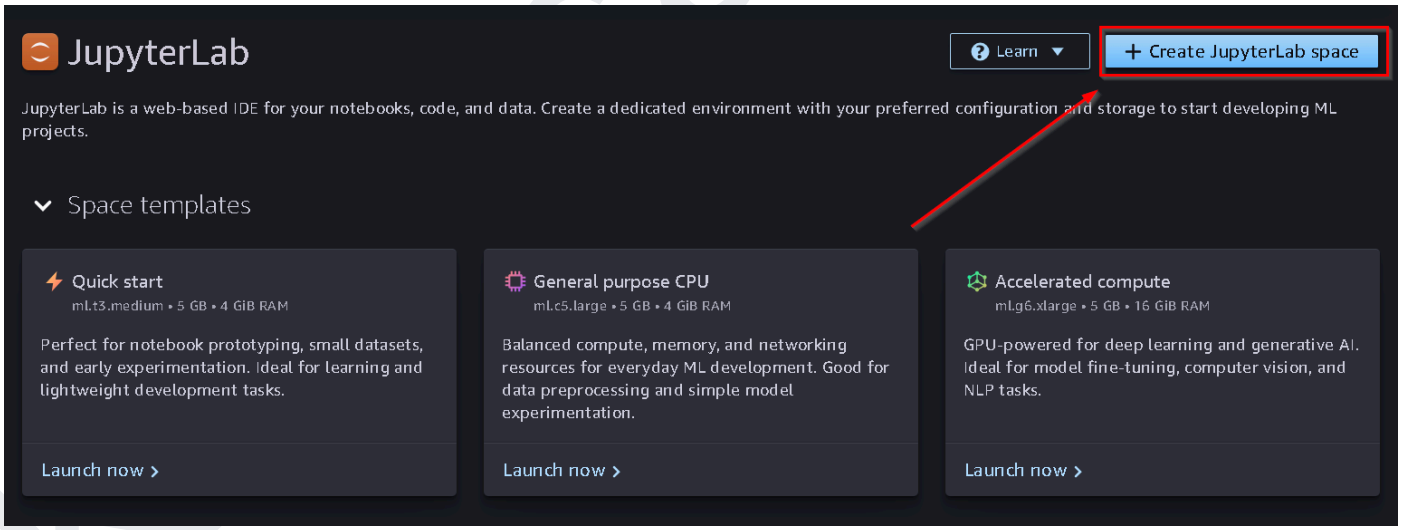
7 SETTING UP JUPYTER NOTEBOOK ENVIRONMENT IN SAGEMAKER STUDIO

In this section, you'll set up and configure a Jupyter Notebook environment in SageMaker Studio. You'll create a lab space and adjust linter settings to ensure a smooth coding experience, preparing your environment for AI and machine learning tasks.

1. Click on the **"JupyterLab"** icon on the left panel.

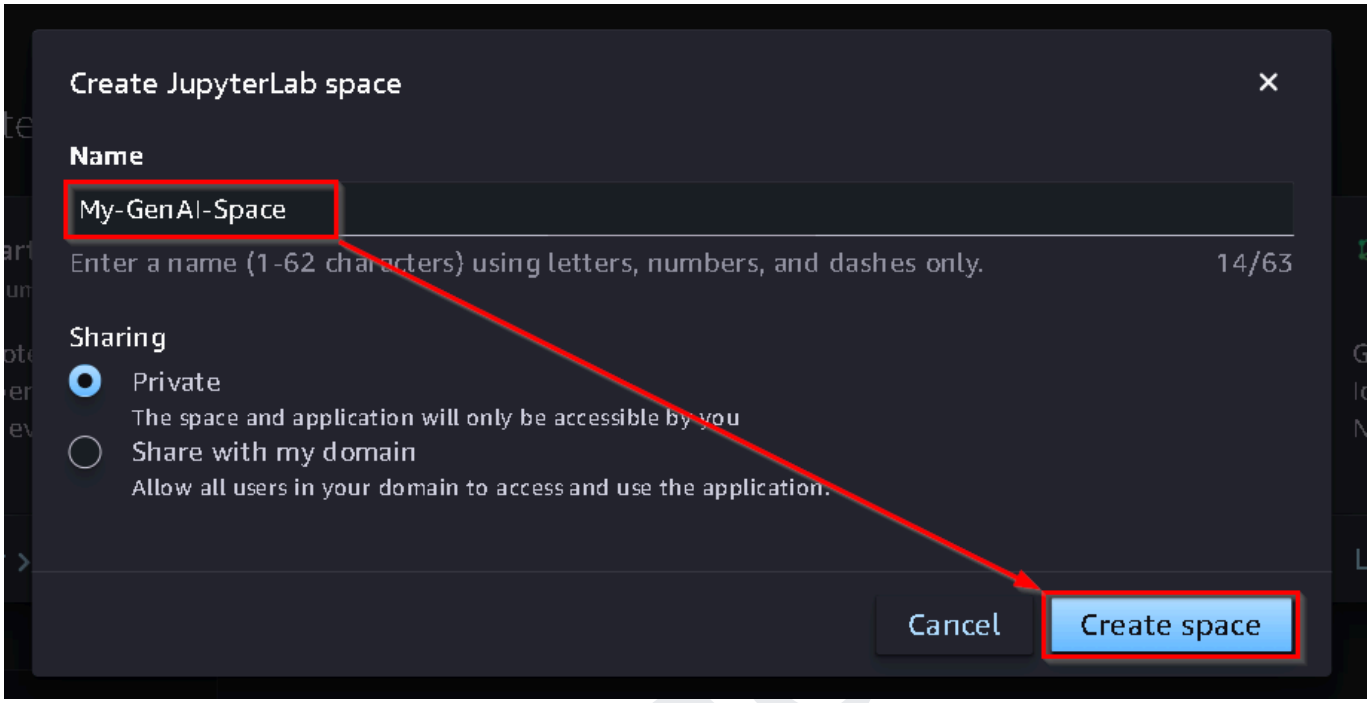


2. Click on the **" + Create JupyterLab Space"** button to create a new workspace.

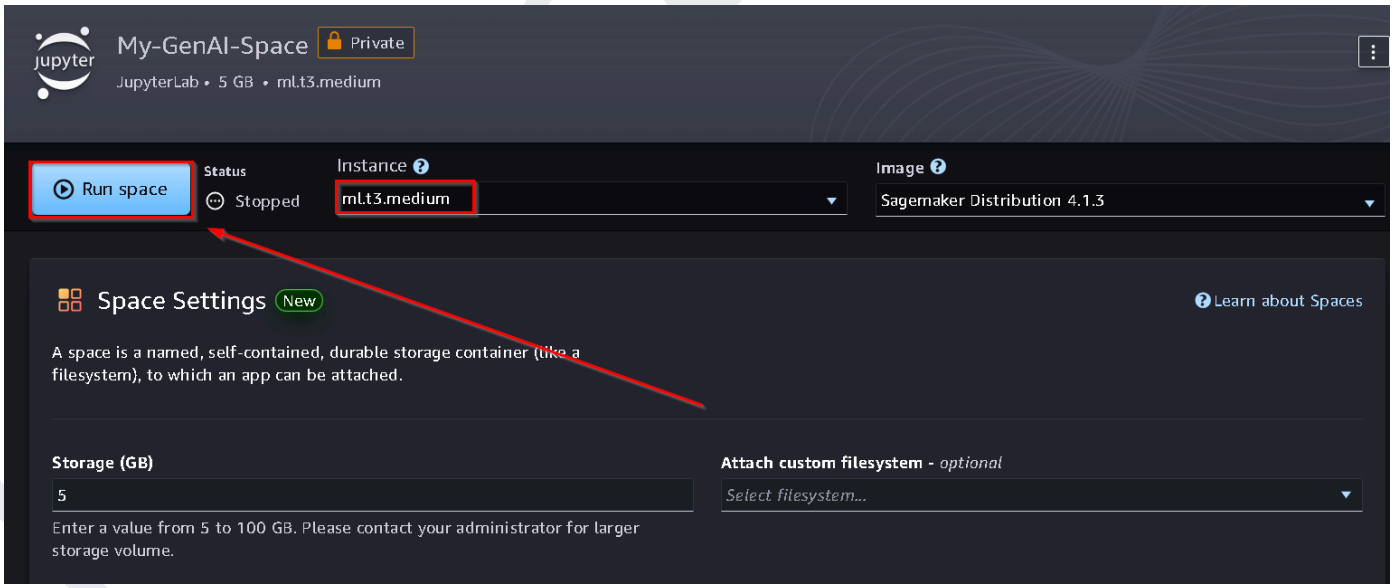


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3. Name the lab space **"My-GenAI-Space."**

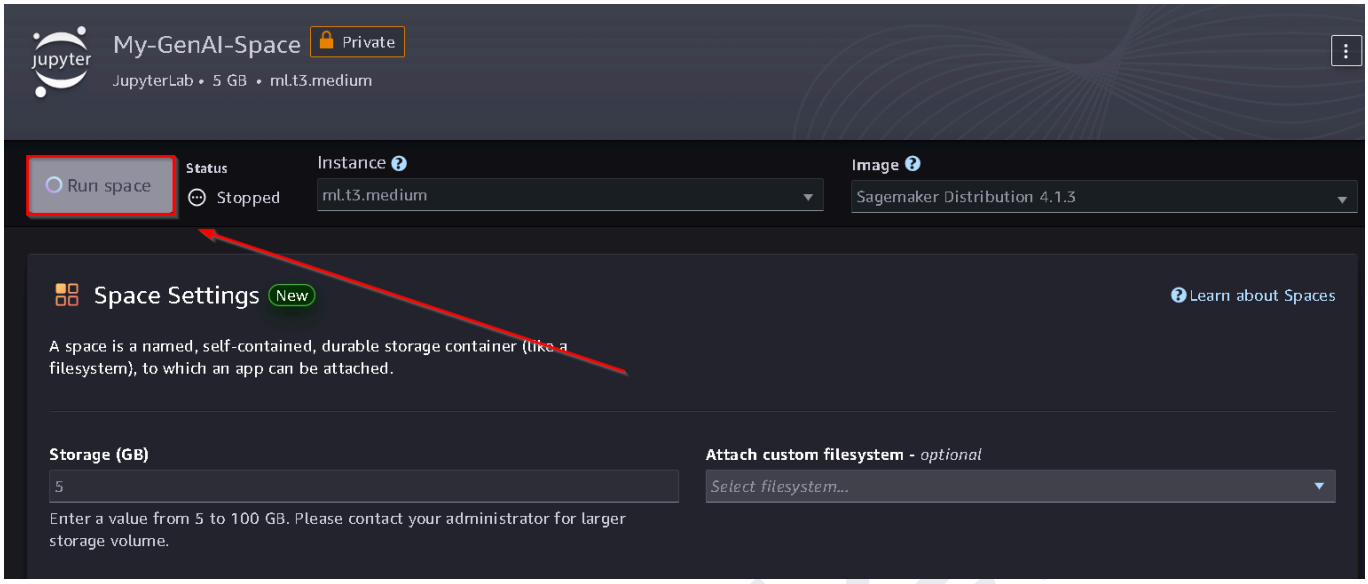


4. Click **"Run space"** and keep all options as **default.**



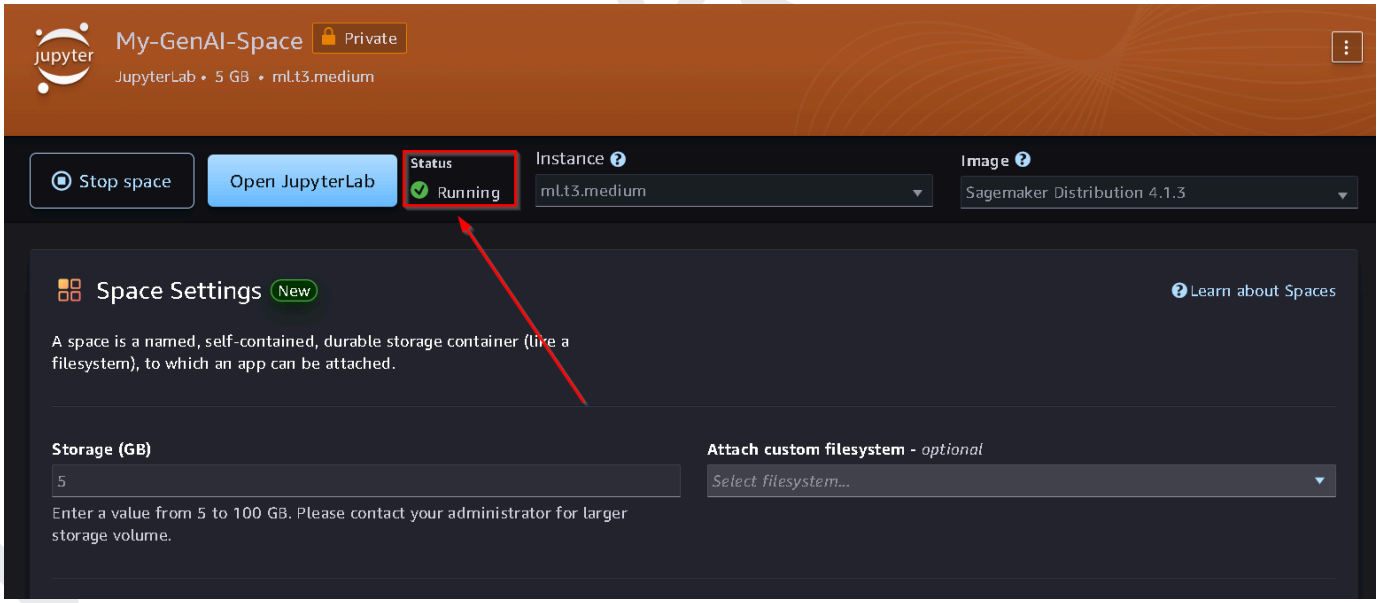
Note: *It may take up to a minute for the space to be created.*

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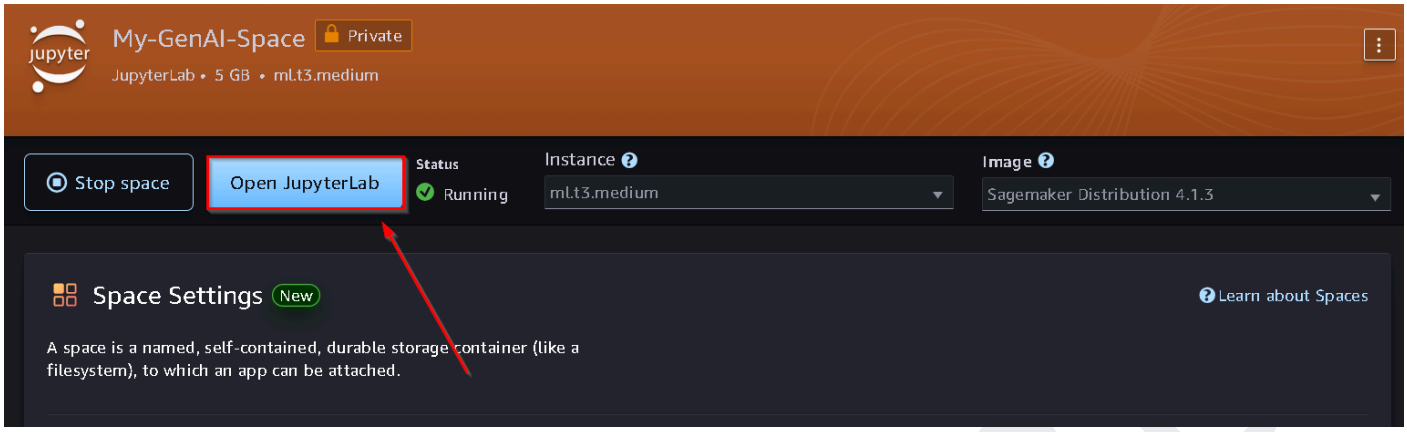
***Note:** The **ml.t3.medium** instance is not free. You will be charged for every second it runs, so ensure you stop the space after completing the lab.*

5. Wait for the lab space to show as **"Running."**

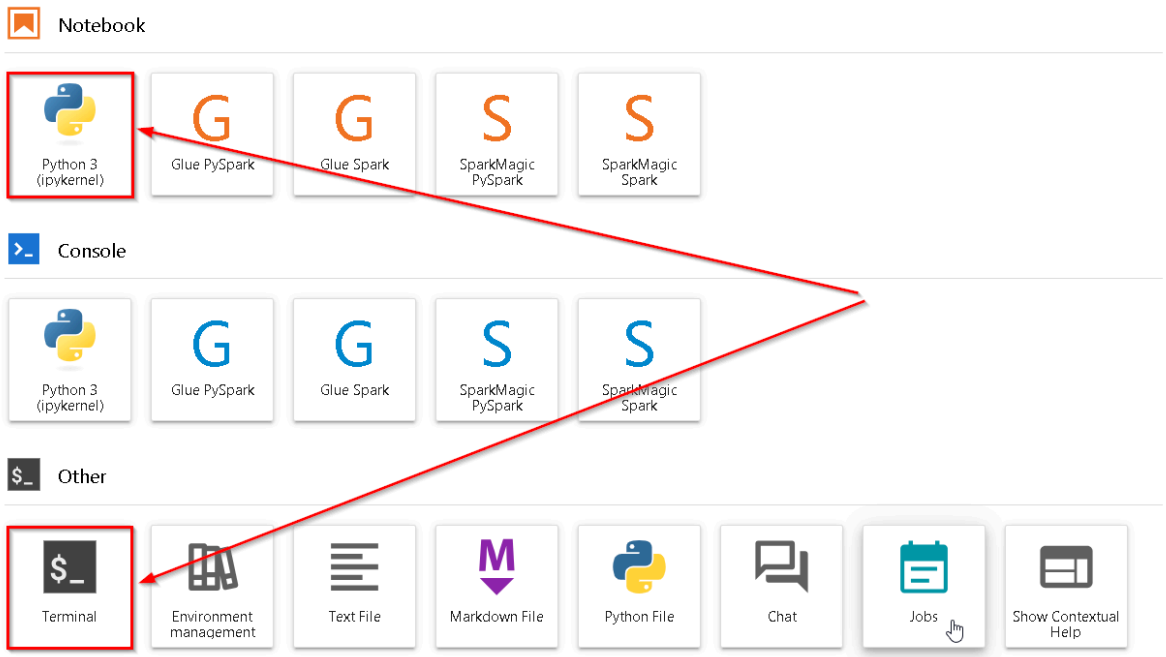


6. Click on **"Open Jupyter Lab now"** to proceed.

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7. Familiarize yourself with the **Jupyter Notebook home screen.**



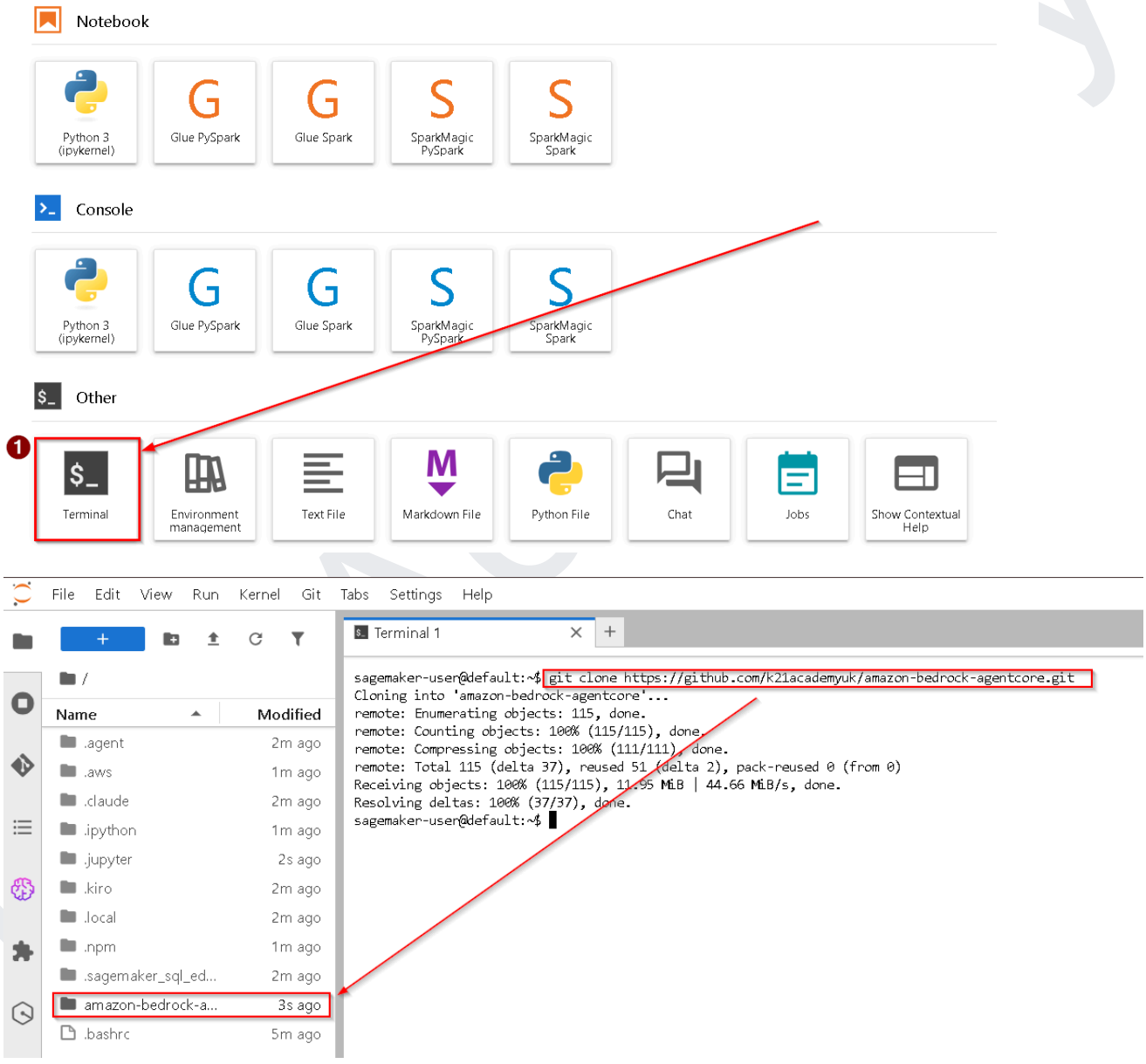
In this workshop, we will primarily be using Python Notebooks and the terminal.

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8 Accessing the AgentCore Workshop Python Notebooks in the Jupyter Environment

1. Open a terminal and clone the Amazon Bedrock AgentCore repository by running the following command:

```
git clone https://github.com/k21academyuk/amazon-bedrock-agentcore.git
```



2. Change to the Amazon Bedrock AgentCore directory.

```
cd amazon-bedrock-agentcore
```

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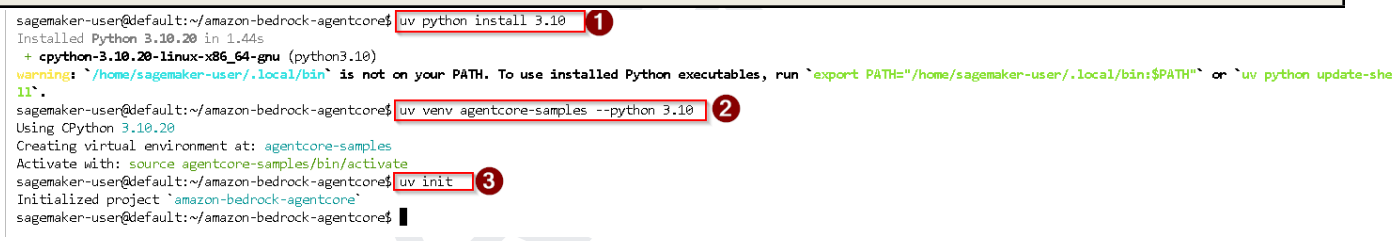
```
sagemaker-user@default:~$ cd amazon-bedrock-agentcore
sagemaker-user@default:~/amazon-bedrock-agentcore$
```

3. To deploy your agent into the AgentCore Runtime, you need to install the necessary prerequisites.

4. For better package management and isolation, it is strongly recommended to create a virtual environment. The uv tool, which is pre-installed in the Jupyter environment, is a high-speed package and project manager for Python.

5. Run the following commands to create and activate the virtual environment:

```
uv python install 3.10
uv venv agentcore-samples --python 3.10
uv init
```



6. Next, add the required packages to your active uv environment:

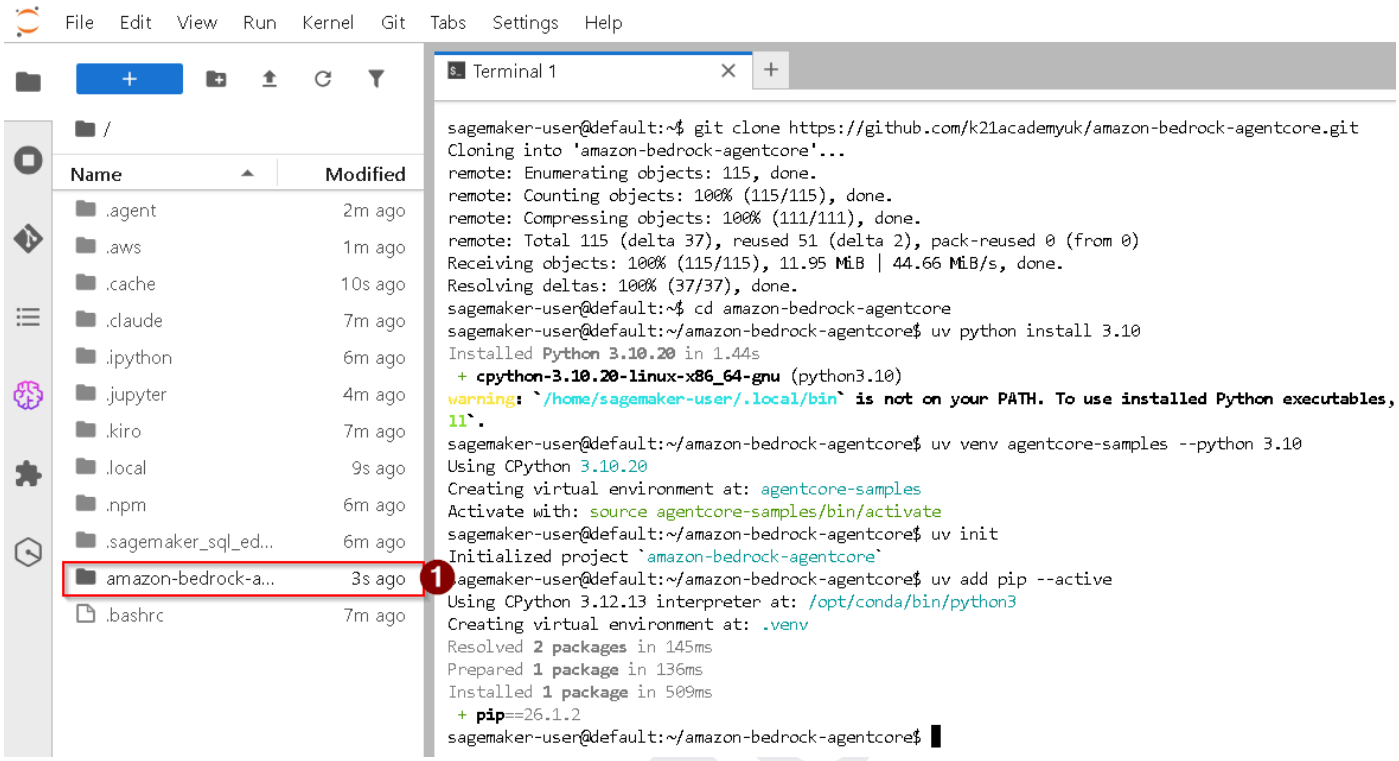
```
uv add pip --active
```

```
sagemaker-user@default:~/amazon-bedrock-agentcore$ uv add pip --active
Using CPython 3.12.13 interpreter at: /opt/conda/bin/python3
Creating virtual environment at: .venv
Resolved 2 packages in 145ms
Prepared 1 package in 136ms
Installed 1 package in 509ms
 + pip==26.1.2
sagemaker-user@default:~/amazon-bedrock-agentcore$
```

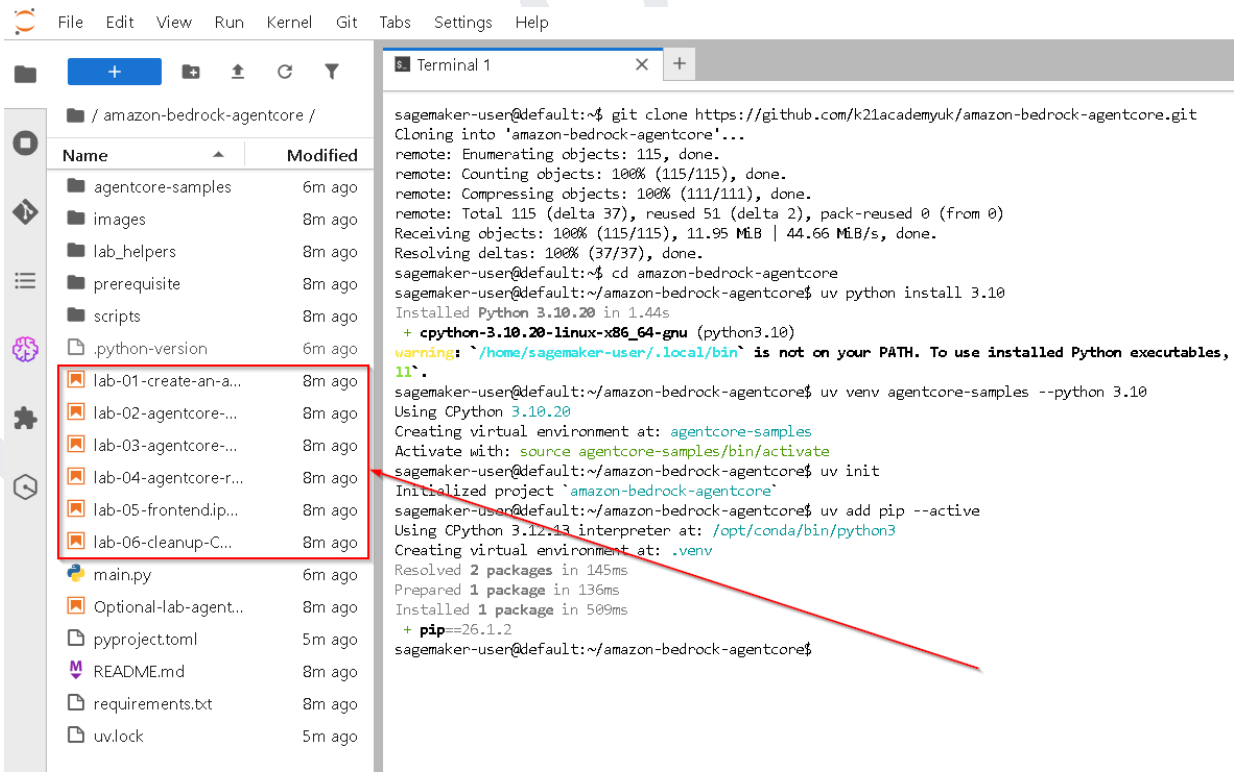
Congratulations! Now, you can start the workshop.

7. The Jupyter notebooks for the end-to-end labs are in the `amazon-bedrock-agentcore` directory.

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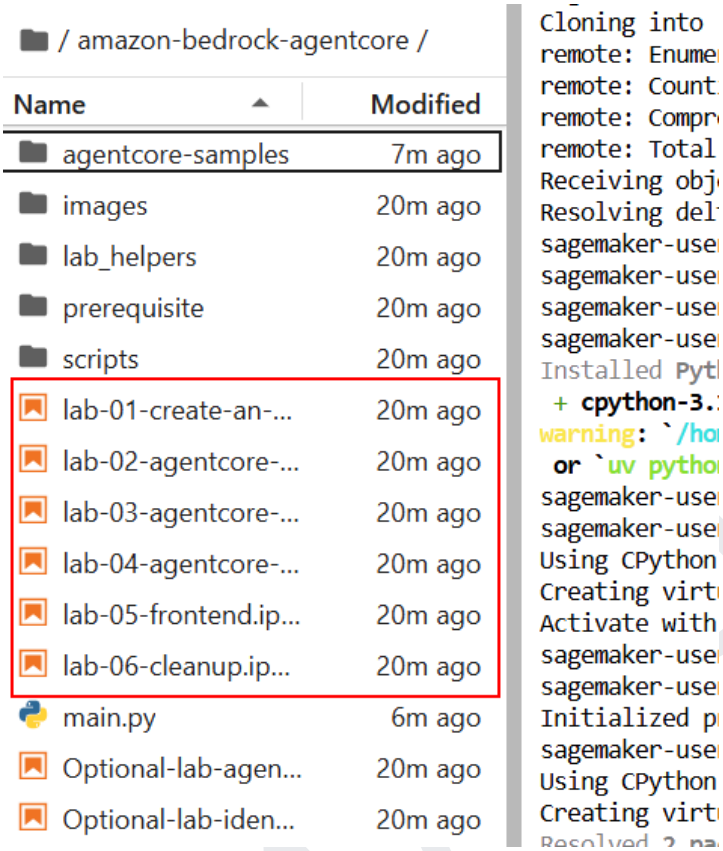
8. Now that the workshop environment is ready, we will work on these **6 Python files** to build production-ready AI agents using Bedrock AgentCore.



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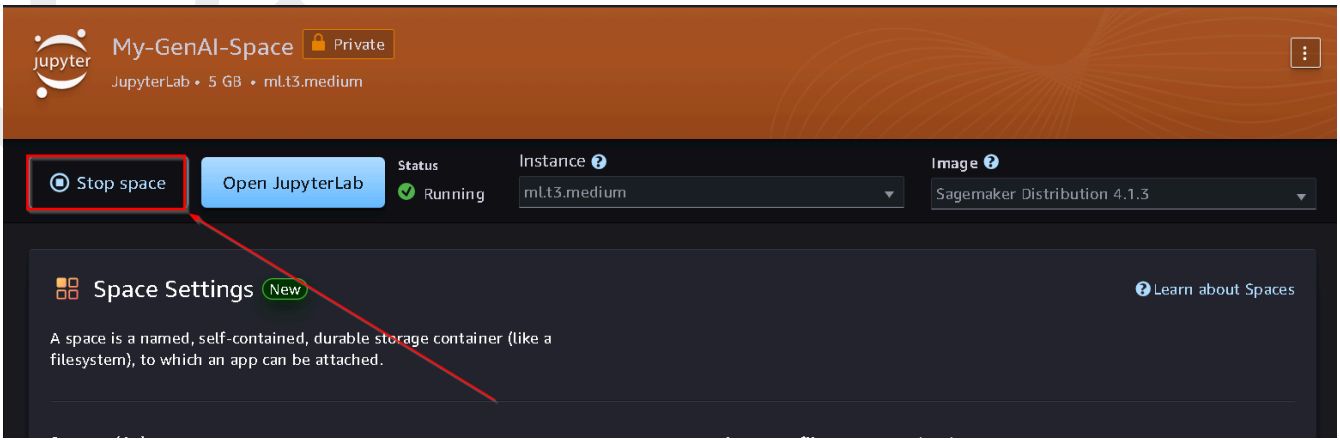
9 Cleanup Resources

Using the first 5 Python files, we will build production-ready AI agents with Bedrock AgentCore. The 6th file, "lab-06-cleanup," will delete all the resources we created. Finally, we will delete the SageMaker domain.

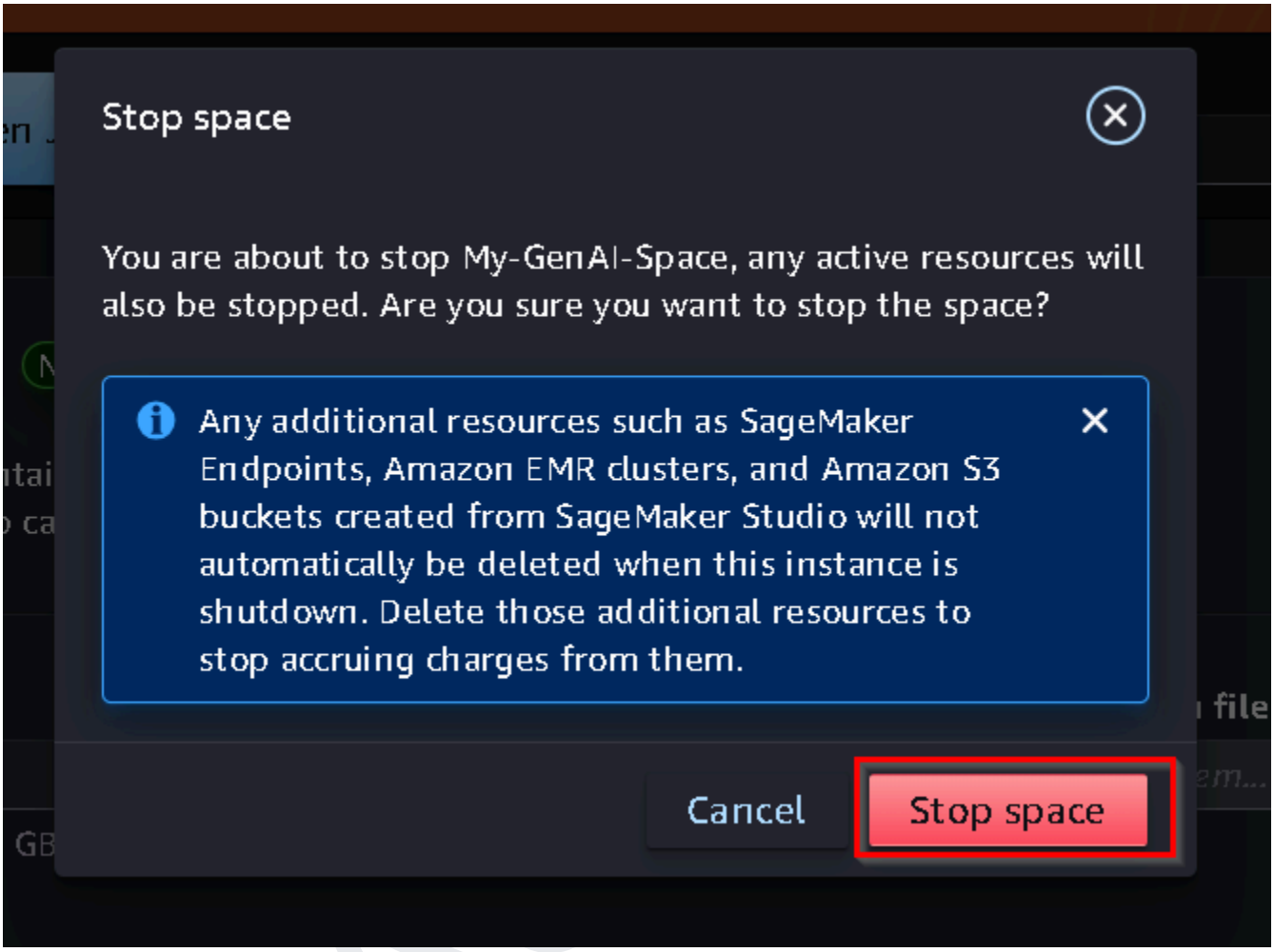


9.1 Delete the Jupyter Space

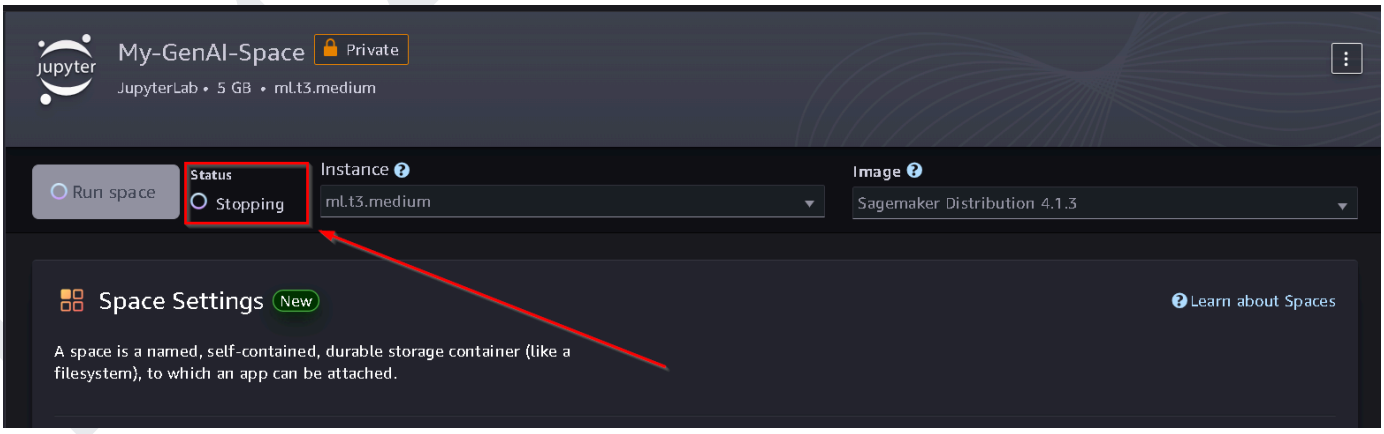
1. Go to the **SageMaker Studio** tab and click on the **Stop Space** button.



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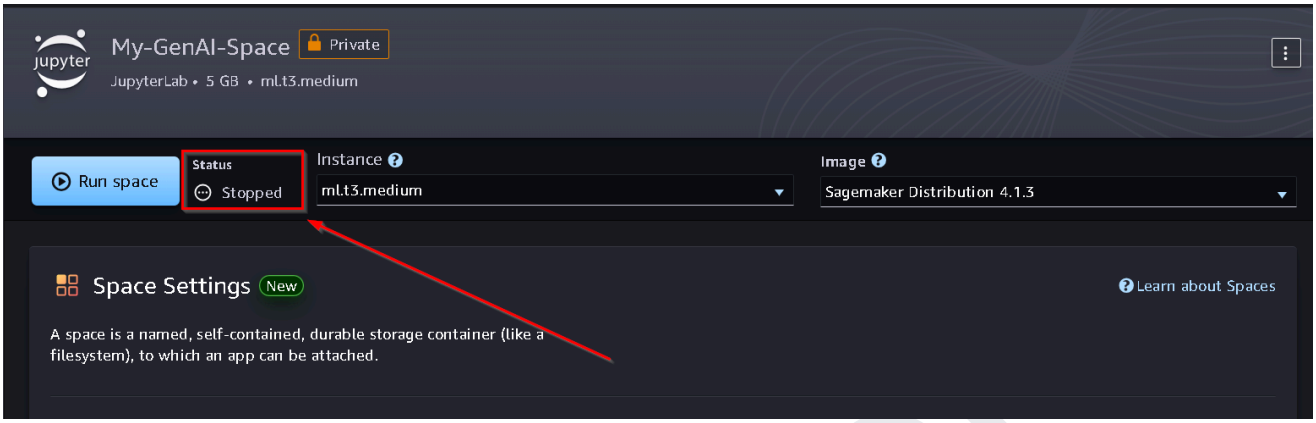
2. The Space is now stopping.



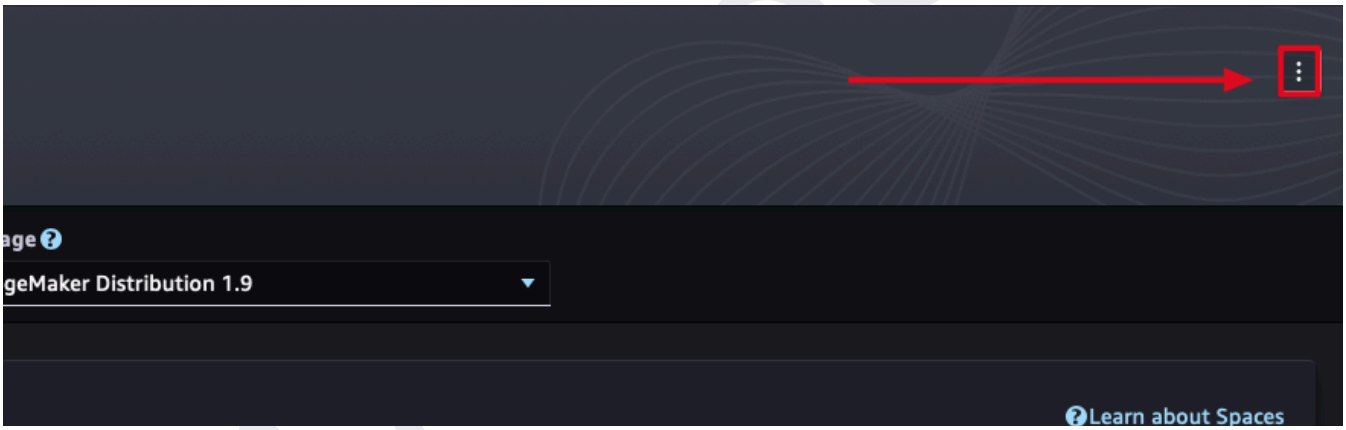
Note: *Wait for the status to show* ***"Stopped"*** *before proceeding.*

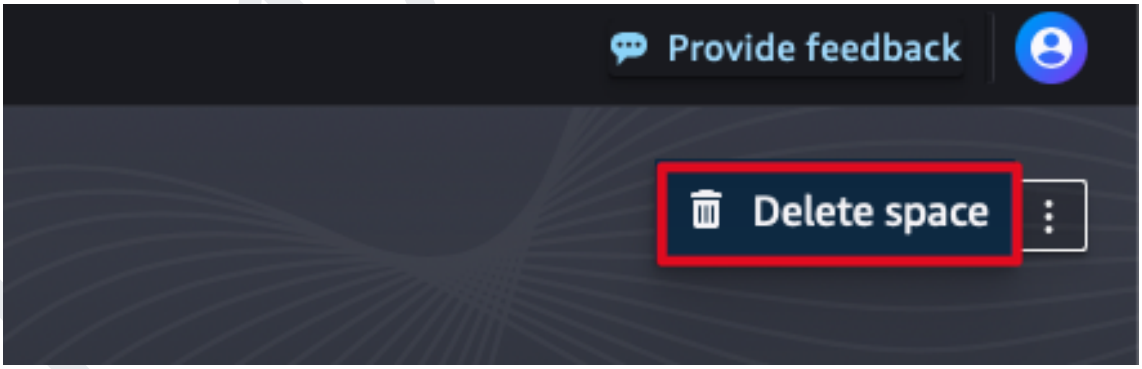
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3. Delete the Lab Space:
   - Check if the space is stopped.

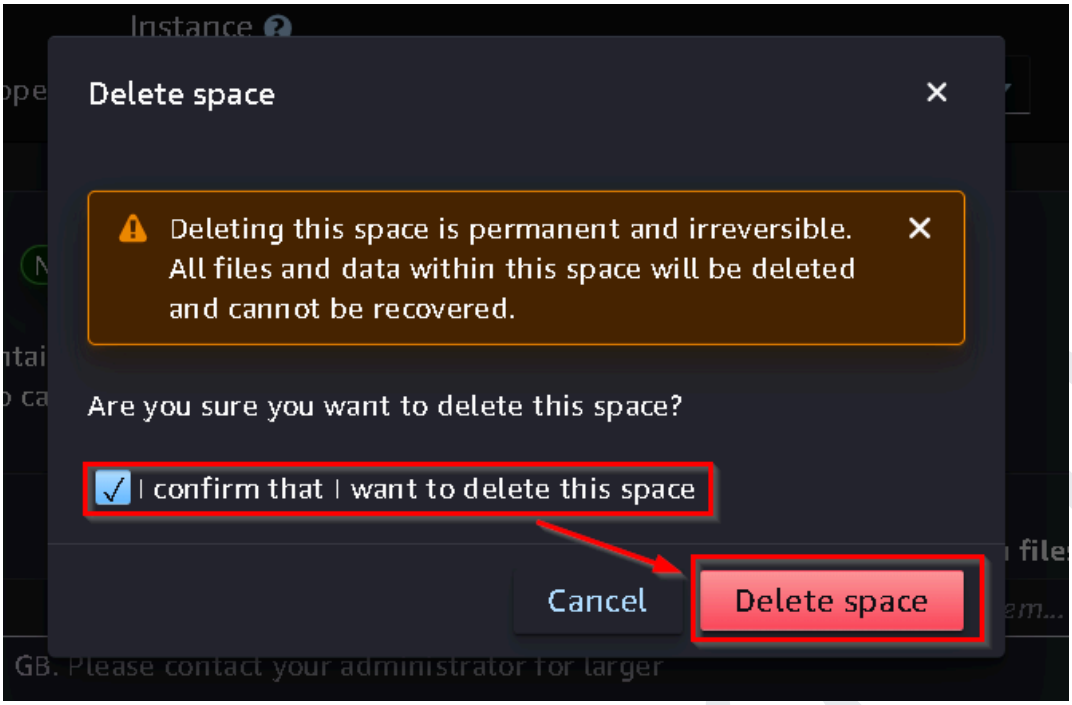


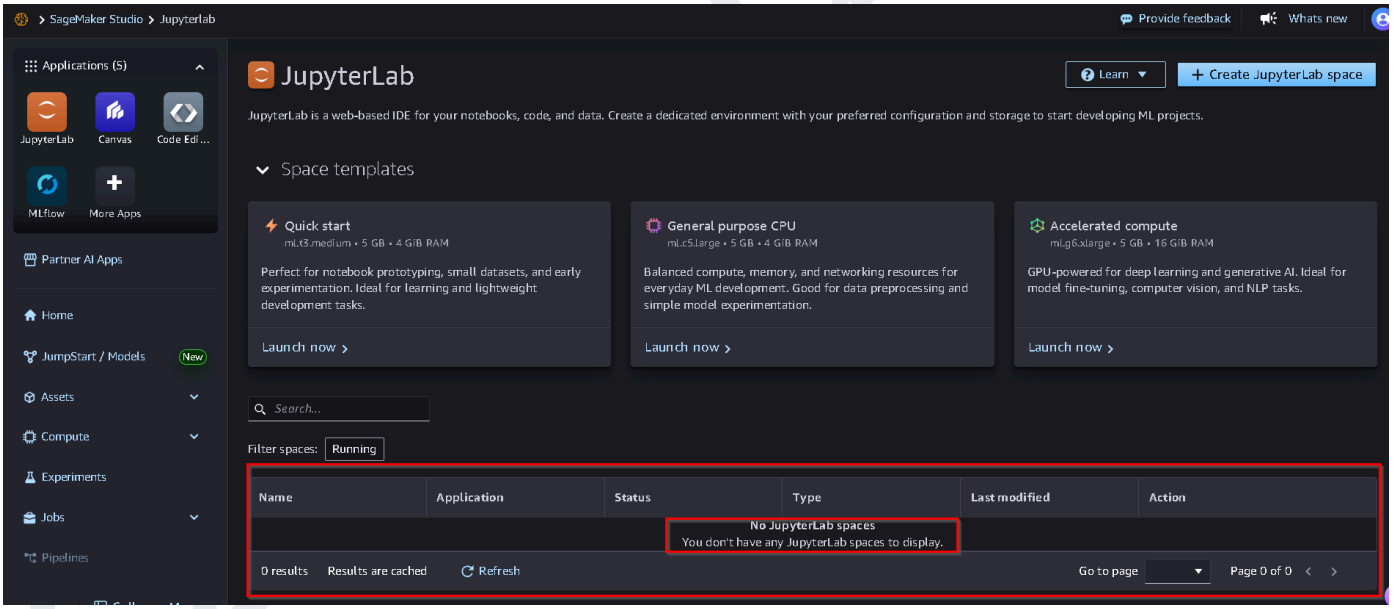
4. Click on the **three dots** on the right side and select **Delete** to remove the space.





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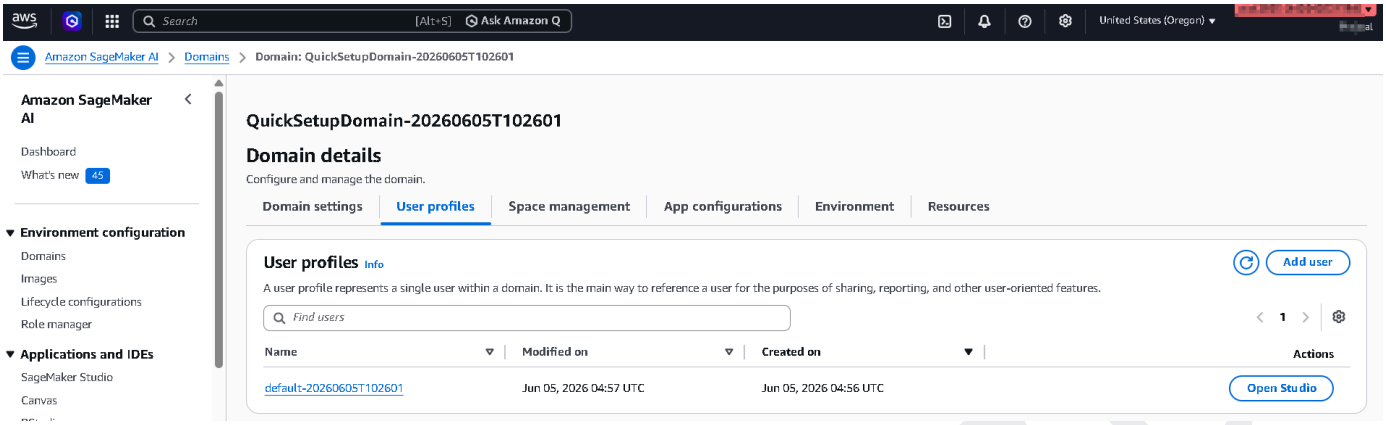


***Note:** After deletion, ensure no Jupyter spaces are listed, indicating the space was successfully deleted.*

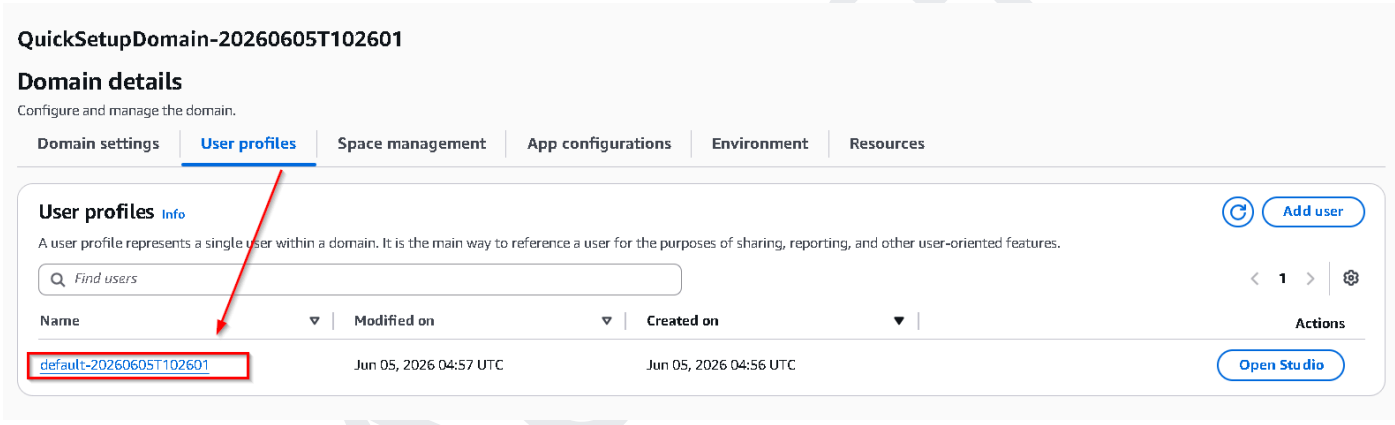
5. Delete the Domain User:
   - Return to the main AWS Console tab.

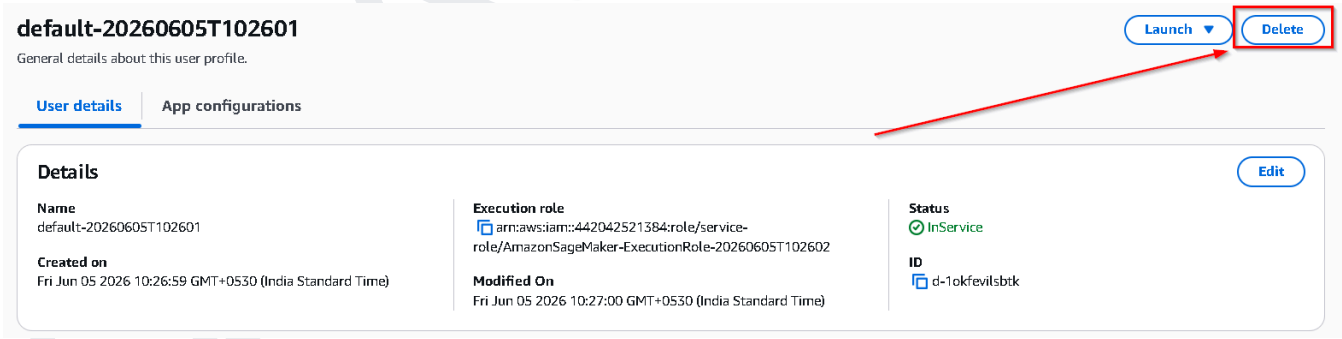
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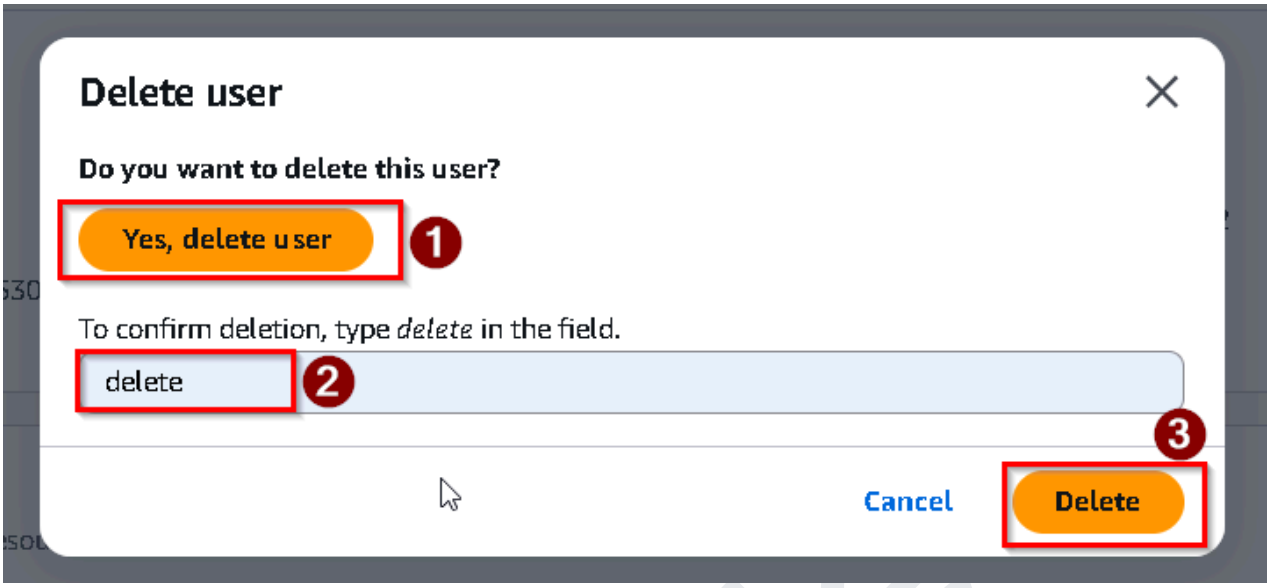


6. Navigate to the **Domain user** section and delete the user that was created for the lab.

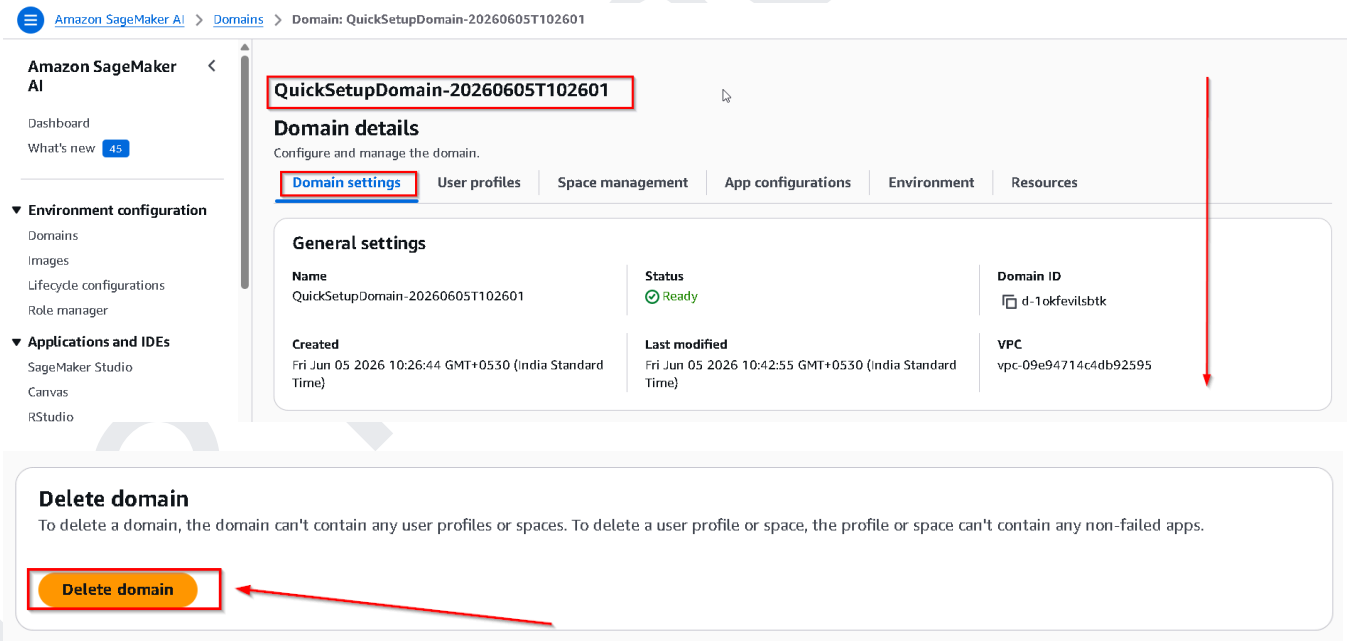




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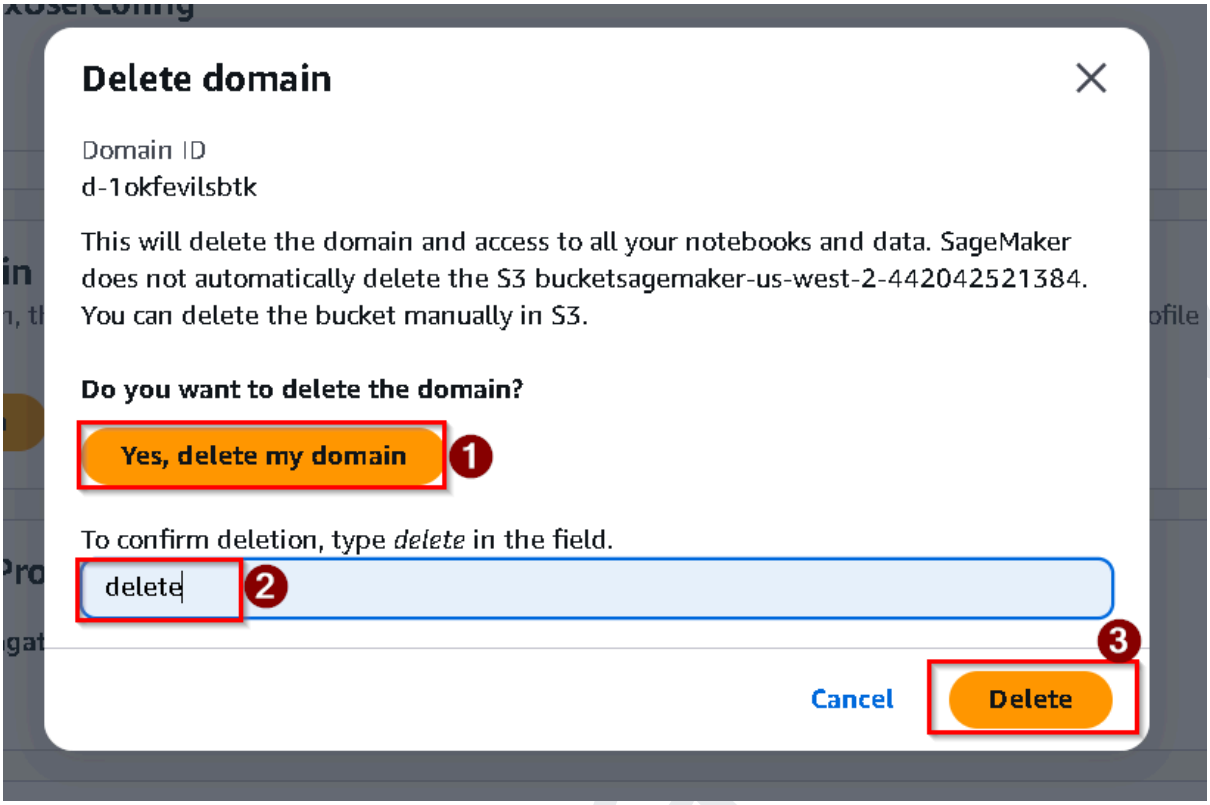


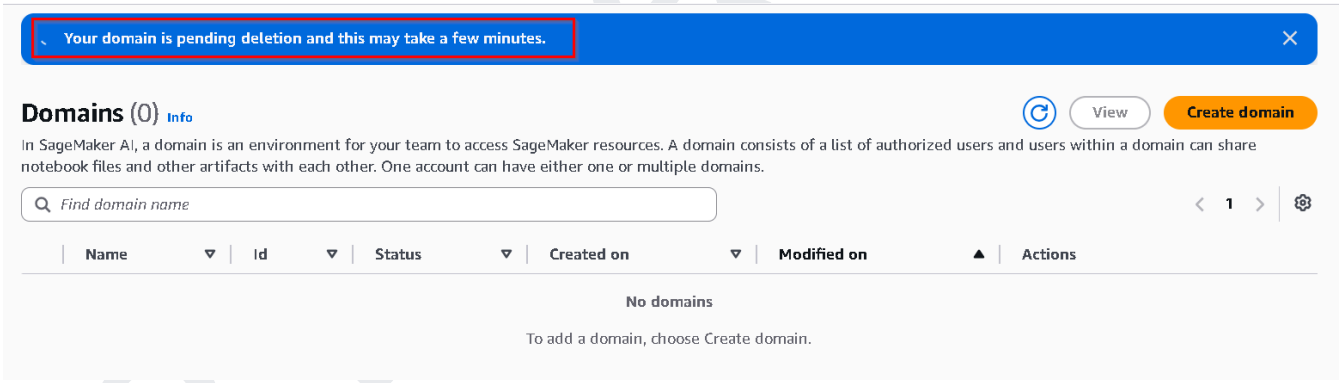
7. Delete the Domain:
   - Go to **domain setting** & scroll down and then click on Delete



- First click on **Yes delete my domain**, then **type delete** & then click **delete**

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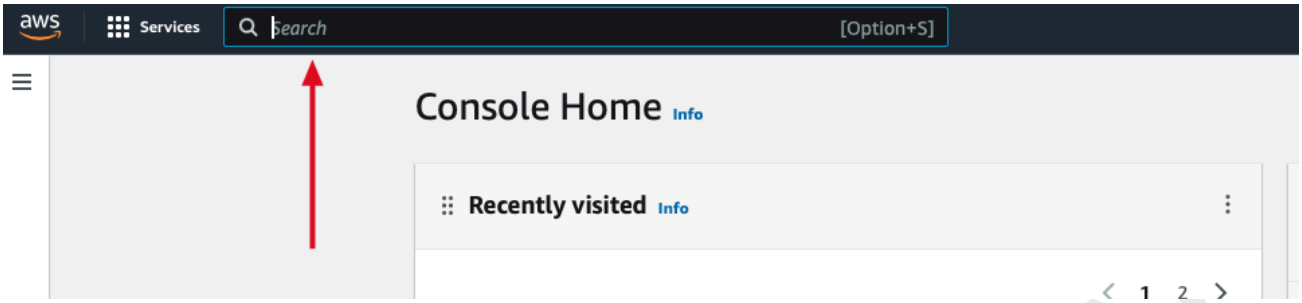


Note: *Wait for 2-3 minutes and refresh the page to confirm the deletion.*

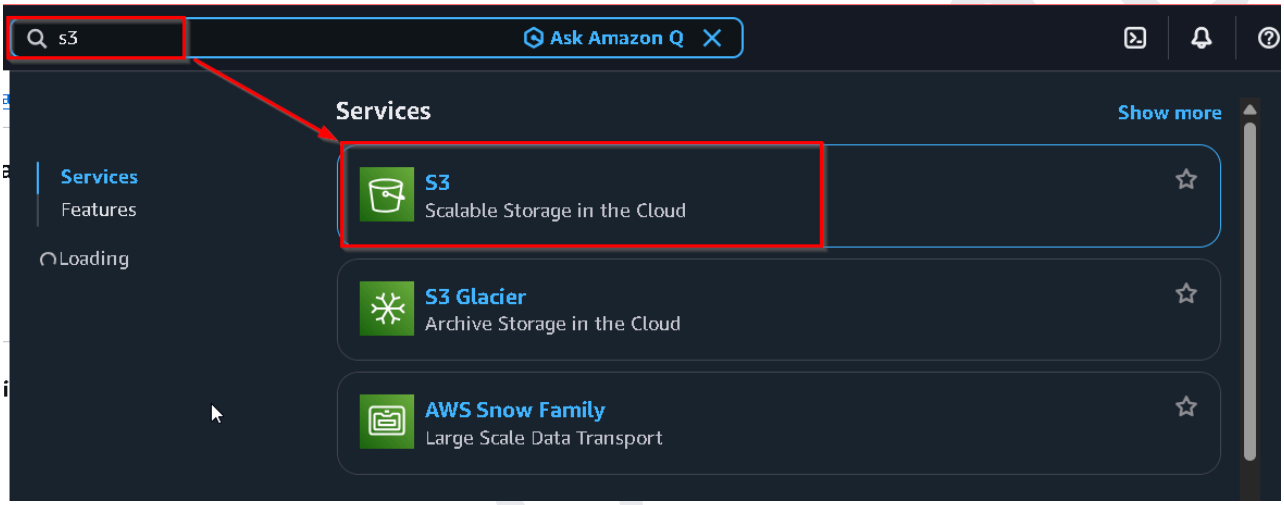
9.2 Delete the S3 Bucket

1. Go to the **AWS Management Console**.

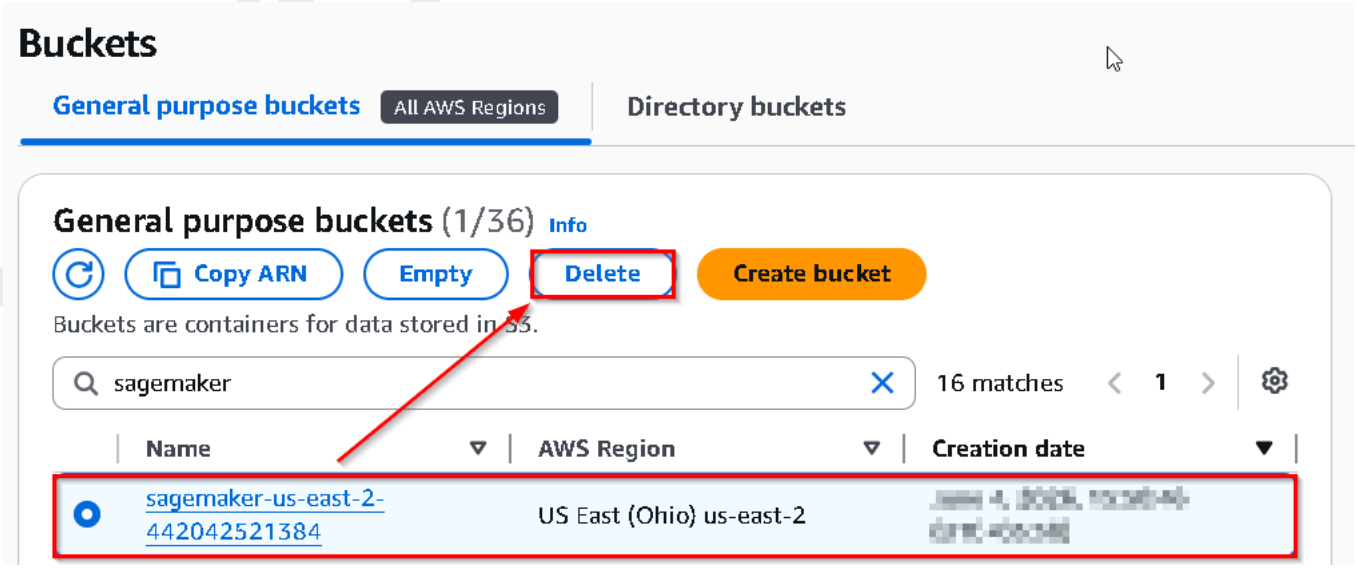
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2. Type **S3** in the search bar and select the **S3** service.

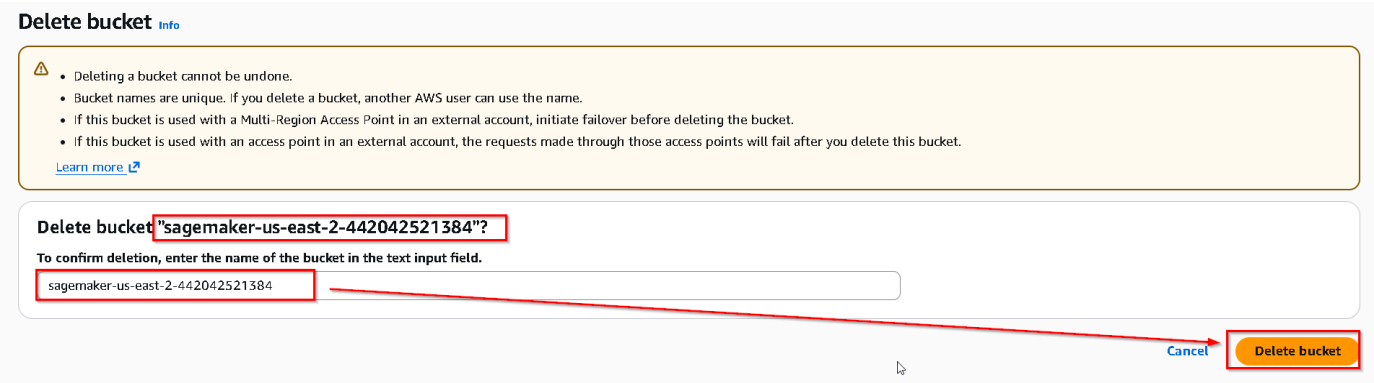


3. Locate the S3 bucket by searching for the domain name created at the start and click on the Delete button.

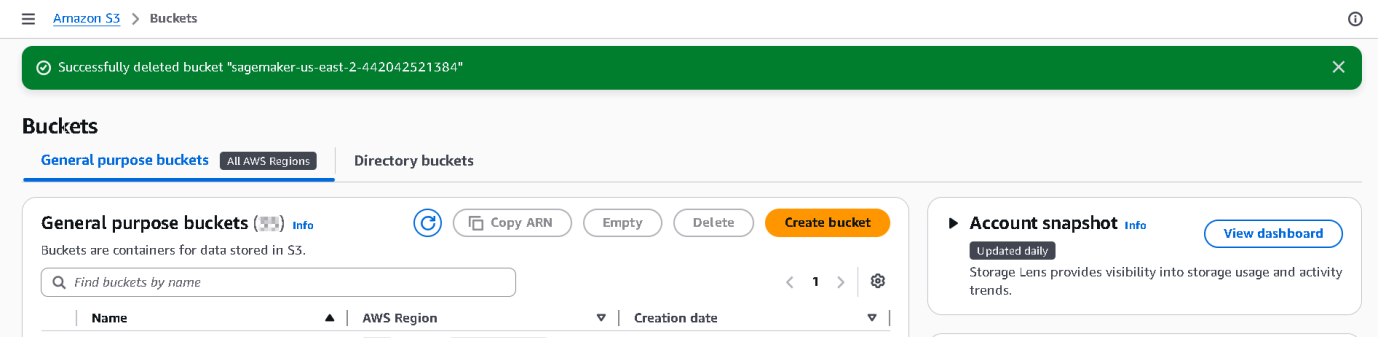


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4. Copy and paste the bucket name in the confirmation text box, and click on the **Delete Bucket** button**.**



5. Verify that the bucket has been successfully deleted.



Note: *By completing these steps, you have successfully deleted all the resources that were provisioned during the domain creation process.*

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10 SUMMARY

In this lab, we have learnt:

- **Creating a SageMaker Studio Domain:** Setting up the foundation for your AI/ML development.
- **Configuring Jupyter Notebooks:** Launching a workspace and optimizing it for coding and data analysis.

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